

Voice and extraction in Malayic

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Abstract

In many Malayic languages (western Austronesian), subjects can undergo A'-movement (relativization, *wh*-movement, focus-movement, and topicalization) without any special qualification, whereas non-subject nominals can only A'-move if the verb appears without a voice prefix. We propose a novel account for the syntax of voice alternations in Malayic languages, synthesizing insights in the prior proposals of Aldridge 2008b and Nomoto 2015, 2021. Adopting the view that the predicate constitutes a *phase* (Chomsky, 2000, 2001), with word order determined at each phase level (Fox and Pesetsky, 2005), our proposal derives both the general subject-only A'-extraction restriction and its limited exception and associated morphological restrictions. The proposal is motivated by our original data on voice and extraction in Suak Mansi Desa, a previously undescribed Malayic language of western Borneo, which we then extend to Standard Indonesian and Standard Malay as well as various other dialects and languages of the region.

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1 Introduction

Austronesian languages are frequently discussed in work in both syntactic typology and theory for their apparent interactions between syntactic movement operations and the choice and form of verbal voice morphology. Within the Malayic subgroup of Austronesian languages, spoken in Indonesia and Malaysia, such interactions have been well-studied in Standard Indonesian and Malay (SI, SM), but much less so among non-standard, regional and colloquial varieties. In this paper, we offer a new analytic perspective on these morphosyntactic interactions, informed by our original fieldwork on Suak Mansi Desa, a previously undescribed Malayic language of western Borneo. Unique properties of the grammar of Suak Mansi Desa suggest that such interactions receive a natural explanation given certain independently-motivated assumptions about the syntactic organization of predicates and how and when morphological forms and word order are determined during syntactic structure-building. Our case study thereby advocates for a particular view of the relationship between morphology and syntax, while also highlighting the important contribution that the description of non-standard language varieties can play in advancing linguistic theory.

We begin by introducing the basics of Malayic voice, concentrating first on the well-studied behaviors of SI and SM. Malayic languages are frequently described as exhibiting an active/passive voice alternation. In SI and SM, active verbs bear a *meN-* prefix, where *N* is a homorganic nasal, and passive verbs bear the prefix *di-*; see (1a,b).¹ Additionally, there is a third clause type, which we refer to as a *bare passive*.² In bare passives, the subject is an internal argument but there is no voice prefix and the agent must be immediately preverbal; see (1c).

(1) Active, passive, and bare passive in SI/SM:

- a. Fera **men-**ulis surat itu. (*meN-tulis* > *menulis*)

Fera ACT-write letter DEM

‘Fera wrote that letter.’

- b. Surat itu **di-**tulis (oleh Fera).

letter DEM PASS-write by Fera

‘That letter was written by Fera.’

- c. Surat itu saya tulis.

letter DEM 1sg write

‘I wrote that letter.’

(SI; based on Sneddon, 1996: xxiii)

¹ We follow the Leipzig glossing conventions, with the addition of MID = middle, MOD = modal. The active voice prefix is glossed ACT except in Desa, where we differentiate between two forms, N and meN.

² The construction has been variously described in the literature as “passive type 2” (Dardjowidjojo, 1978; Sneddon, 1996, a.o.), “bare passive” (Voskuil 1996, 2000; Cole et al. 1999b; Nomoto 2021, a.o.), or “object(ive) voice” (Arka and Manning, 1998; Cole et al., 2008; Legate, 2012a, 2014; Jeoung, 2018b), among other terms; see Nomoto 2006.

From each of these basic clause types in (1), the subject is the only nominal that can undergo A'-movement³ without further qualification, leading to the common description of these languages as exhibiting a *subject-only restriction* on movement. See Keenan 1972 and Keenan and Comrie 1977 as well as Kroeger 1991 for important, earlier descriptions of Austronesian languages as exhibiting such a restriction.

There is, however, an important exception to the subject-only restriction, in structures that we refer to as *object extraction* and exemplified in (2) below. Such examples have been described by Chung (1976a: 74) and Cole and Hermon (1998, 2000) as extraction of an object from an active clause such as (1a) but with the *meN-* active prefix undergoing “deletion” as a response to A'-movement across it. Note that these examples cannot simply be described as involving A'-extraction of a bare passive subject, for reasons we review in section 2.3 below. See also Saddy 1991, Soh 1998, Aldridge 2008b, and Sato 2012 among others for prior discussion of this effect.

(2) **Object extraction requires verb without *meN-* active prefix in SI/SM:**

Buku [_{RC} yang Wati { ***men**-ulis / tulis } ____] ada di atas meja itu.
 book C Wati ACT-write write exist on top table that
 ‘The book that Wati wrote is on the table.’ (SI; Cole and Hermon, 2005: 67)

The so-called “*meN*-deletion” observed in (2) and its relationship to the bare passive as in (1c), as well as the more general question of the nature of subject-only extraction restrictions and their exceptions, have received substantial attention in previous theoretical work on these languages.

In this paper, we advance a new approach to the syntax and verbal morphology of Malayic languages, which serves to explain the observed link between verbal forms and restrictions on movement, as well as the attested range of variation in these behaviors in the region. Empirically, we first introduce and analyze the voice alternation in (Suak Mansi) Desa, a previously undescribed Malayic language of West Kalimantan, Indonesia, as particularly informative regarding the nature of the relationship between voice morphology and the extraction restriction.⁴ Active voice forms in Desa are notable for having two variants: a short nasal prefix of *N-* alone and a full, long *meN-* prefix, which are generally in free variation. The long prefix *meN-* disallows object A'-movement, as in SI/SM, but the short prefix *N-* allows it:

³ We use the term *A'-movement* as a cover term for *wh*-movement, relativization, focus-fronting (clefting), and topicalization. A'-movement stands in opposition to *A-movement*, which for example includes the movement of the subject from its thematic base position to its clause-initial position, as discussed in section 3.1. For expository purposes, in sections 1 and 2, only gaps of A'-movement are indicated, as _____. We also use the terms “extraction” and “movement” interchangeably.

⁴ There is no standardized orthography for Desa, so we utilize a modified SI/SM orthography. Following SI/SM orthography, nasals are represented by <ng> [ŋ] and <ny> [ɲ]. We use <'> for [ʔ].

(3) **Desa non-subject A'-movement possible with short *N-* but not with long *meN-* prefix:**

Opai yang inya { **m-ewa'** / ***mem-ewa'** } ____? (*N-bewa'* > *mewa'*)
what C 3sg N-bring *meN-*bring
'What did s/he bring?'

Desa also has a bare passive with immediately preverbal agent, parallel to the SI/SM example (1c). The verb in Desa bare passives as in (4) has no voice prefix, disallowing both the short (*N-*) and long (*meN-*) active prefixes.

(4) **Desa bare passive verb allows neither active prefix:**

Kayu inya { *bewa'* / ***m-ewa'** / ***mem-ewa'** }.
wood 3sg bring N-bring *meN-*bring
'S/he is bringing wood.'

We argue that these facts from Desa offer several important hints for the analysis of Malayic voice, including the nature of the *meN-*deletion interaction in the standard languages. Desa teaches us that, in a Malayic language with greater flexibility in the form of active voice verbs, object extraction still restricts the form of the verb as in SI and SM, but that it is specifically the initial syllabic *me-* whose occurrence is restricted in object extraction, as the short nasal prefix of *N-* alone remains, as in (3). At the same time, the availability of *N-* in object extractions such as (3) underscores the position that such clauses cannot be straightforwardly analyzed as underlyingly bare passive clauses like (4).

The analysis we propose has the following notable properties. First, we treat the *meN-* of Malayic active verbs as reflecting two affixes, *me-* and *N-*, in Desa as well as other languages of the subgroup (informed by suggestions in Benjamin 2009 and Gil 2002). We propose that these two affixes correspond to two different functional heads, following a proposal for Malay verb phrase structure from Nomoto 2015, 2021. Second, we adopt the view that the verbal domain constitutes a derivational cycle: a *phase* in our framework of analysis (see Chomsky, 2000, 2001). Once the verbal phase is built, the phonological realization and word order of its contents are determined (Fox and Pesetsky, 2005 et seq); although further syntactic operations may target the predicate and its contents, they must not violate the word order established at the predicate level. With these assumptions in place, we are able to derive the almost exceptionless subject-only restriction on nominal A'-movement in these languages, and the role that overt morphological form plays in its one systematic exception. This contrasts from previous accounts for these languages which do not offer any deeper explanation for the relationship between the extraction restriction and the morphological form of the verb.

An additional strength of our proposal is that it naturally extends to and explains grammatical behavior across a broader range of languages of the region. There is substantial variation in the form and behavior

of the voices across the many regional, colloquial Malay varieties and other Malayic languages (Gil, 2002; Adelaar, 2005c; Donohue, 2008; Nomoto, 2020; a.o.) but — with a few notable exceptions such as in work by Cole, Hermon, and colleagues (2006; 2008) — the consequences of this variation for the theoretical analysis of these voice forms have not been substantially explored. We first advance our proposal for Desa, but then demonstrate how it naturally extends to the behavior of Standard Indonesian and Malay, as well as various other dialects and languages of the region such as Kendayan and Madurese.

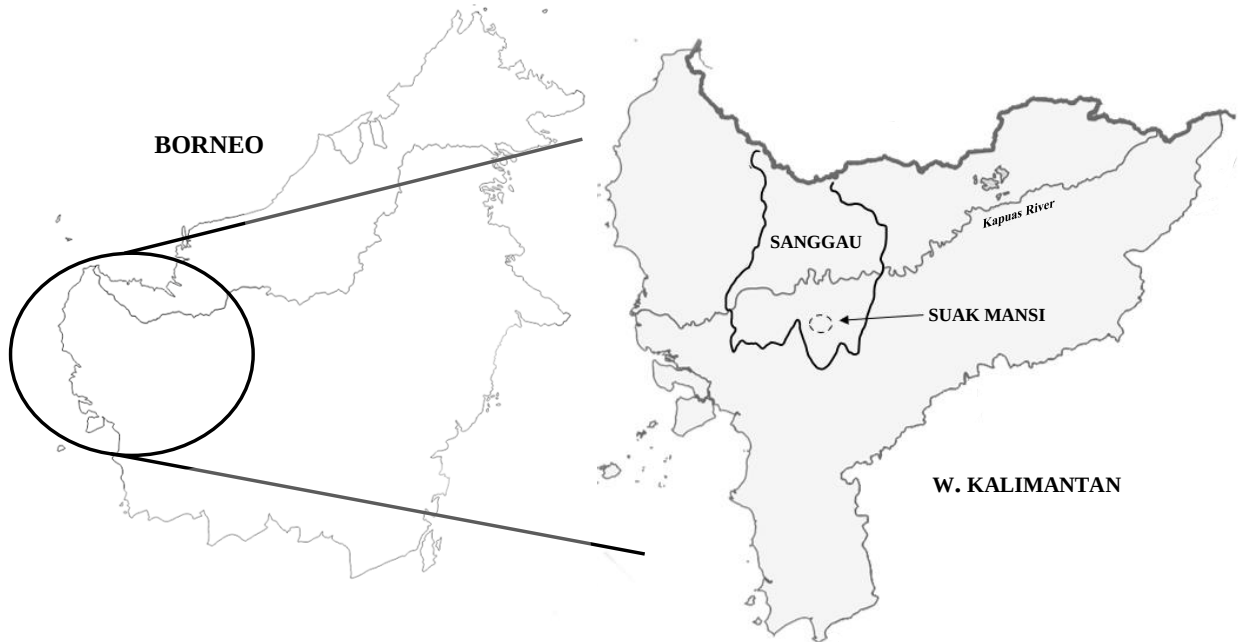
The paper is structured as follows. First, in section 2, we offer a description of voice alternations, A'-extraction, and their interaction in (Suak Mansi) Desa, a previously undescribed Malayic language of western Borneo. There we also introduce the parallel facts in Standard Indonesian and Standard Malay, which have been well described and theorized in prior work, highlighting the few but important differences between them and Desa. Next, in section 3, we present our novel proposal for Malayic clausal syntax and how it accounts for the interaction of A'-extraction and voice morphology in Desa. Then, in section 4, we extend our analysis to the standard languages, deriving the “*meN*-deletion” interaction from the properties of linearization in a phase-based syntax. We discuss some details of verbal morphosyntax and its variation across Malayic varieties and beyond in section 5, which offers further support for our account. We conclude in section 6.

2 Voice and extraction in Desa

2.1 Language background

Desa is a Malayic language spoken in West Kalimantan, Indonesia, in western Borneo.⁵ The variety discussed here is predominantly spoken in a village called Suak Mansi, which is located south of the Kapuas River in the Sanggau Regency. The approximate location of Suak Mansi is given below in Figure ?? . Speakers of this language refer to it as *bahasa Desa* (‘Desa language’) and occasionally colloquially as *Opai* (the Desa word for ‘what’).

⁵ Desa belongs to the West Bornean Malayic subbranch within the Smith 2017 classification of Malayic, and therefore not a variety of Malay proper. We caution that there is another, distinct language that is also called Desa and spoken further upriver (Collins, 2004; Herpanus, 2014; Collins and Herpanus, 2018), but based on the available data, it differs considerably from the language discussed here. Where it is necessary to clearly distinguish the two, we suggest referring to the language discussed here as Suak Mansi Desa. See Sommerlot 2020: 8–13 for discussion of Suak Mansi Desa in relation to the Desa described by Collins and Herpanus as well as its position within the Malayic family.



All Desa data in this paper were collected through primary fieldwork by the second author over the course of three summers beginning in 2017. Elicitation sessions were conducted in the village of Sungai Galing, which is located just north of Suak Mansi, along the Kapuas River. All data from these sessions have been transcribed and tagged and are publicly available in the Kaipuleohone Digital Language Archive (Sommerlot, 2023). Unfortunately, due to their remote nature, access to both Sungai Galing and Suak Mansi is quite difficult. As a result, we are limited here to the data collected previously and reflected in the archive. The behavior that we report here reflects consistent contrasts across multiple sentences with different lexicalizations, over multiple sessions, which faithfully reflects the data in our entire corpus. In the few instances where there are conflicting judgments or insufficient evidence to address a question in our corpus, we explicitly note this.

2.2 Voices

Desa generally resembles well-studied Malayic languages in terms of its word order and voice marking. As described earlier, Standard Indonesian and Malay (SI/SM) have active and passive voices with corresponding verbal prefixes (*meN-* and *di-*), as well as a so-called “bare passive” where the verb appears without a voice prefix. Desa patterns closely with SI/SM in also having these three types of clauses. Here we describe these three clause types in Desa in turn, highlighting salient differences from the behavior of SI/SM. We concentrate here on the behavior of eventive bivalent predicates.

Active An active, transitive construction in Desa is subject-initial and its verb is prefixed with a nasal affix. Objects follow the verb, and neither the agent nor the theme are marked for case. The construction in (5) with the prefix *meN-* strongly resembles the Standard Indonesian example provided earlier in (1a). However, there is also another variant of this same sentence, identical in every way except that the nasal prefix has been “shortened” to only the homorganic nasal. Both forms of the nasal prefix occur frequently on a variety of verbs, and there is no distinction in meaning between the two variants.⁶

(5) **Desa active voice with *N-* and *meN-*:**

Inya { **ng-**ikat / **meng-**ikat } perau yetn.
 3sg N-tie ME_N-tie boat DEM
 ‘S/he ties the boat.’

The two forms behave identically in their nasal realization: *N-*, alone as the short prefix and as part of the long prefix *meN-*, undergoes homorganic nasal substitution if the verb stem is consonant-initial, and otherwise appears as *ng-*.⁷ We present a number of transitive verbs in their bare, *N-*-prefixed, and *meN-*-prefixed forms in (6), organized by the stems’ onset type.

(6) **Some Desa transitive verbs with their *N-* and *meN-* active forms:**

<i>pangkong</i>	<i>mangkong</i>	<i>memangkong</i>	‘hit’
<i>bewa’</i>	<i>mewa’</i>	<i>memewa’</i>	‘bring’
<i>boli</i>	<i>moli</i>	<i>memoli</i>	‘buy’
<i>tanam</i>	<i>nanam</i>	<i>menanam</i>	‘plant’
<i>tulis</i>	<i>nulis</i>	<i>menulis</i>	‘write’
<i>kirim</i>	<i>ngirim</i>	<i>mengirim</i>	‘send’
<i>sapah</i>	<i>nyapah</i>	<i>menyapah</i>	‘call’
<i>cuci</i>	<i>nyuci</i>	<i>menyuci</i>	‘wash’
<i>ikat</i>	<i>ngikat</i>	<i>mengikat</i>	‘tie’
<i>ambe’</i>	<i>ngambe’</i>	<i>mengambe’</i>	‘take’

⁶ As a reviewer notes, Soh and Nomoto (2009, 2015) claim that there is a difference in Malay between *meN-*-prefixed active verbs and so-called “bare actives” (see note 20 below), in the range of predicates they take and their aspectual interpretation (cf Jeoung 2018a: 87). Kaswanti Purwo (1986) also reports related contrasts in Indonesian. Efforts were made to see if there are any such differences between *N-* and *meN-*-prefixed active forms in Desa, or any restrictions on their cooccurrence with different temporal and aspectual auxiliaries, but no such differences were found.

⁷ Our field notes unfortunately include very few liquid-initial transitive stems, with only *nge-*-prefixed active forms attested. For instance, one such stem is *liat* ‘see’, with attested active form *ngeliat*. We suspect that *ngeliat* is a short form, but the predicted long form *mengeliat* was not tested.

In active clauses with preverbal auxiliaries, the subject must precede the auxiliaries. This is shown for Desa in (7) with a range of preverbal auxiliaries, with both short (*N-*) and long (*meN-*) voice prefix verbs. This generalization is also true of SI/SM, as seen in (8). We analyze this word order as reflecting obligatory movement of the subject to a high structural position.

(7) **Agent subjects precede auxiliaries in Desa active voice:**

Aku { *tongah* / *udah* / *nda'* / *mau'* / *tau'* } { *ny-apah* / *meny-apah* } *kawan-ku*.
 1sg PROG ASP NEG FUT MOD N-call MEN-call friend-1sg
 'I {am calling / called / didn't call / will call / can call} my friend.' (*N-sapah* > *nyapah*)

(8) **Agent subjects precede auxiliaries in SI/SM active voice:**

{*Kami*} *tidak akan* {**kami*} *mem-baca buku ini*.
 1pl.EXCL NEG FUT ACT-read book DEM
 'We will not read this book.' (SI; Cole et al., 2008: 1512)

Di- passive The *di-* passive in Desa is nearly identical to what has often been referred to as the 'canonical' passive in SI/SM: the verb bears a *di-* passive voice prefix and the agent can be specified with an optional prepositional phrase headed by *oleh* 'by.'

(9) **Desa canonical passive:**

Makanan mau' di-kirim (*oleh uma'-ku*) *ke Meliau*.
 food FUT PASS-send by mother-1sg to Meliau
 'Food will be sent to Meliau (by my mother).'

Bare passive Desa additionally has a "bare passive" clause type. This construction is characterized by a verb without any voice prefix, an internal argument as the clause-initial subject, and an immediately preverbal external argument (agent): see (10). We italicize these immediately preverbal agents throughout this section. This construction shares several features in common with the bare passive in SI/SM, as in (11).

(10) **Desa bare passive:**

Meja yetn aku teipel.
 table DEM 1sg touch
 'I touch the table.'

(11) **SI/SM bare passive:**

Mobil itu dapat kita perbaiki.
 car DEM MOD 1pl.INCL repair
 'We can repair the car.'

(Chung, 1976b: 60)

The bare passive in SI/SM is so called as its verb must appear without any voice prefix. The same is true of the bare passive in Desa: in particular, both short (*N-*) and long (*meN-*) form active prefixes are disallowed, as seen in (12). The *di-* prefix is also disallowed in the bare passive; see (13).

(12) **Desa bare passive verb allows neither active prefix: =(4)**

Kayu *inya* { bewa' / ***m**-ewa' / ***mem**-ewa' }.
 wood 3sg bring N-bring MEN-bring
 'S/he is bringing wood.'

(13) **Desa bare passive verb disallows the *di-* prefix:**

Mano' *uma'-ku* { buat / ***di**-buat }.
 chicken mother-1sg make PASS-make
 'My mother makes chicken.'

Two notable properties of the agent in bare passives are also the same between SI/SM and Desa. First, the agent of a bare passive is obligatory, unlike the agent of a *di-* passive:

(14) **Agents are obligatory in Desa bare passives:**

- | | |
|--|--|
| <p>a. Buku *(<i>ikau</i>) beca.
 book 2sg read
 'You read the/a book.'</p> | <p>b. Buku yetn udah { di-tulis / *tulis }.
 book DEM ASP PASS-write write
 'The book was written.'</p> |
|--|--|

(15) **Agents are obligatory in SI/SM bare passives:**

Mobil ini akan *(*kita*) perbaiki.
 car DEM FUT 1pl.INCL repair
 'We will repair this car.'

(based on Chung, 1976b: 85)

Second, in clauses with preverbal auxiliaries, the agent must be in immediately preverbal position, following the auxiliaries. This is shown with the progressive auxiliary in (16). The same holds of bare passive agents in SI/SM (see e.g. Dardjowidjojo, 1978: 259), as in (17). This placement of bare passive agents is in striking contrast to the agentive subjects of active clauses, which must precede any auxiliaries; see examples (7–8) above.

(16) **Agents are low in Desa bare passives:**

Buku yetn { **lelaki yetn* } tengah { *lelaki yetn* } tulis.
 book DEM PROG male DEM write
 'The man is writing the book.'

(17) **Agents are low in SI/SM bare passives:**

Buku ini {**kami*} tidak {**kami*} akan {*kami*} baca.
book DEM NEG FUT 1pl.EXCL read

‘We will not read this book.’

(SI; Cole and Hermon, 2005: 62–63)

Bare passive clauses therefore cannot be simply described as active clauses with clause-initial object topics. These two types of constructions differ in their verb forms and in the position of their agents with respect to auxiliaries, in a correlated fashion, as seen in (18):

(18) **Bare passive versus active clause with object topicalization in Desa:**

a. Surat to mau’ *kame*’ { kirim / **ng-irim*⁸ } ke Ali.
letter DEM FUT 1pl.EXCL send N-send to Ali

‘We will send the letter to Ali.’

b. Surat to, *kame*’ mau’ *ng-irim* ___ ke Ali.
letter DEM 1pl.EXCL FUT N-send to Ali

‘The letter, we will send to Ali.’

Additionally, if a bare passive agent is pronominal, it can be reduced and procliticize to the verb. For example, the first singular pronoun *aku* appears as *ku* in (19) and (20) below.

(19) **Pronominal agents can procliticize to the verb in Desa bare passives:**

Yetn [ular [RC yang ___ suwah *ku-liet* ___] paling bosa].
DEM snake C AUX 1sg-see most big

‘That is [the biggest snake [that I’ve ever seen]].’

(20) **Pronominal agents can procliticize to the verb in SI/SM bare passives:**

Engkau akan *ku-beli-kan* piano.
2sg FUT 1sg-buy-APPL piano

‘I will buy you a piano.’

(SI; MacDonald and Darjowidjojo, 1967: 243)

The one point where the bare passive in Desa diverges from that in SI is in the types of nominals that are allowed as its agent. Bare passive agents in SI are limited to pronouns like the first-plural *kita* in (11) above or various pronoun substitutes (Sneddon, 1996: 248–249; Kaufman, 2014; a.o.). Desa does not have this restriction, so agents in the bare passive can be full noun phrases (DPs), as seen in (16) above. However, Desa is far from unique within the Malayic branch in this flexibility of bare passive agent form. Nomoto (2021) shows that many languages of western Indonesia — including SM (in contrast to SI) as well as many

⁸ We also predict the long form *mengirim* to be ungrammatical here, but this was not tested in this specific example.

non-standard, regional Malay varieties and related languages — allow non-pronominal agents in the bare passive. Against this broader typology, the restriction in SI is in fact the exceptional extreme.

2.3 A'-extraction constructions

Next we turn to patterns of A'-extraction of nominal arguments in Desa as well as in SI and SM. We comment on non-nominal A'-extraction in section 3.2 below.

As noted in the introduction, the interaction of A'-extraction with the *meN*- active prefix in SI/SM has received much attention in prior theoretical work. Desa shows us a slightly different version of such facts, which will motivate our analysis below. As background, we first present the well-known subject-only extraction restriction in SI/SM and its so-called “*meN*-deletion” exception. A'-extraction can target subjects of active clauses as in (21), as well as subjects of *di*- passives and bare passives (not shown here), with no change to verbal morphology. In contrast, the extraction of non-subject arguments is ungrammatical in the basic case. This has led reference works on SI and SM to describe nominal A'-extraction constructions (e.g. gapped relativization and *wh*-fronting) to be restricted to the subject; see e.g. Sneddon 1996: 285–287, 301–302.

(21) Active subject extraction in SI/SM:

Siapa-kah yang ____ telah **mem**-baca buku itu?

who-Q C ASP ACT-read book this

‘Who has read the book?’

(SM; Soh, 1998: 297)

However, object extraction *is* possible in SI/SM, as long as the active voice prefix is absent, as in (22) below. As Soh (1998), Musgrave (2001), and Cole and Hermon (2005) discuss, some theme A'-extraction examples similar to (22) but without an auxiliary could be analyzed as involving extraction of a bare passive subject, rather than A'-extraction of an active object across the verb. However, examples such as (22) where the agent (*Ali*) precedes the auxiliary (*tidak*) show that object A'-extraction from an active clause — as diagnosed by the high, pre-auxiliary position of the agent — is indeed possible, as long as the active prefix is absent.

(22) Object extraction is not possible with *meN*- but possible without, in SI/SM:

baju-baju [_{RC} yang Ali tidak { ***mem**-basuh / basuh } ____]

shirt-RED C Ali NEG ACT-wash wash

‘clothes that Ali isn’t washing’

(SM; Keenan, 1972: 183–184)

This phenomenon of so-called “*meN*-deletion” has received substantial attention in the theoretical literature on SI and SM, for two reasons. First, it shows that the general subject-only restriction on nominal

A'-extraction — a widespread typological trait that holds of many western Austronesian languages (see e.g. Himmelmann 2005: 161–163, Chung and Polinsky 2009: 665–666) — is not absolute in SI and SM. Second, it suggests that verbal voice morphology interacts with A'-extraction in non-trivial ways.

With this background on nominal A'-extraction in SI/SM in place, we now turn to the corresponding behaviors in Desa. Subject A'-extraction from both active and passive clauses are common and unrestricted; see (23–24). Notice that subjects of active clauses with both the short (*N-*) and long (*meN-*) active prefix can be extracted.

(23) **Desa active subject extraction:**

- | | |
|---|--|
| <p>a. Sopai yang ___ ng-ikat perau yetn?
 who C N-tie boat DEM
 ‘Who tied the boat?’</p> | <p>b. Sopai yang ___ meng-ambe’ ageng-ku?
 who C MEN-take basket-1sg
 ‘Who took my basket?’</p> |
|---|--|

(24) **Desa *di-* passive subject extraction:**

- Opai yang ___ tau’ di-tanam di taman?
 what C MOD PASS-plant in field
 ‘What can be planted in the field?’

Next we turn to the A'-extraction of active objects in Desa, which exhibits a slight variation on the “*meN*-deletion” interaction of SI/SM summarized above. While the *meN-* active prefix is disallowed in object A'-extraction just as in SI/SM, no such restriction exists for the shorter *N-* prefix, as seen in the object focus-fronting construction in (25).

(25) **Object extraction allows *N-* but not *meN-* in Desa:**

- Buku to yang opa’-ku { **m**-oli / ***mem**-oli } _____. (*N-boli* > *moli*)
 book DEM C father-1sg N-buy MEN-buy
 ‘It’s this book that my father bought.’

In addition, Desa differs from SI/SM in the word order of object extraction clauses such as (25). While the SI/SM object relative clause in (22) reflects the word order found in active clauses, in Desa object extraction, agents are always immediately preverbal, following any auxiliaries. Example (26) shows the ungrammaticality of the agent in pre-auxiliary position. The examples in (27) below are additional object extraction examples which illustrate the same word order with immediately preverbal agent, with a range of different auxiliaries. Examples (27b–d) also explicitly illustrate the grammaticality of *N*-prefixed verbs but ungrammaticality of

meN-prefixed forms in these structures.⁹ We are not aware of any other Malayic variety which patterns with Desa in the position of agents in object extraction clauses.

(26) **Agents follow auxiliaries in Desa object extraction:**

Opai yang { **inya* } nda' { *inya* } m-ilau ____? (*N-pilau* > *milau*)

what C NEG 3sg N-look.for

'What isn't s/he looking for?'

(27) a. Opai yang { tengah / udah / nda' / mau' / tau' / nda' tau' } opa'-ku m-oli ____?

what C PROG ASP NEG FUT MOD NEG MOD father-1sg N-buy

'What is my father buying?' / 'What { did/didn't/will/can/can't } my father buy?'

b. Opai yang udah *inya* { ng-ikat / *meng-ikat } ____?

what C ASP 3sg N-tie MEN-tie

'What did will s/he tie?'

c. Opai yang nda' *inya* { m-ewa' / *mem-ewa' } ____? (*N-bewa'* > *mewa'*)

what C NEG 3sg N-bring MEN-bring

'What didn't s/he bring?'

d. Opai yang tengah *inya* { n-anam / *men-anam } ____? (*N-tanam* > *nanam*)

what C PROG 3sg N-plant MEN-plant

'What is s/he planting?'

As we noted earlier, this "Aux ag V" order is found in bare passives, but not in active clauses. However, at the same time, recall that verbs in bare passives cannot be prefixed with either version of the active prefix; see (12). We conclude that object extraction clauses in Desa as in (25–27), with their "Aux ag V" order and *N*-prefixed verbs, are neither straightforward active clauses nor straightforward bare passive clauses. The analysis of this object extraction clause type in Desa, and its relation to the SI/SM *meN*-deletion pattern above, will thus feature prominently in our analysis below.

2.4 Summary

We conclude this section with a summary of the different clause types involving eventive bivalent verbs in Desa. We give a schematic illustration of each clause type and their morphological and word order characteristics in (28). Aux* indicates the position where any and all preverbal auxiliaries such as negation, modals, and tense/aspect markers appear.

⁹ One object extraction example with the agent preceding the auxiliary was accepted in a grammaticality judgment task, but this word order was later rejected as ungrammatical when checked in a later session. Such structures were never accepted otherwise or naturally produced.

(28) **Summary of Desa clause types:**

a. Active:	subj/ag	Aux*	(<i>me</i>) <i>N</i> -	V	obj/th
b. <i>Di</i> - passive:	subj/th	Aux*	<i>di</i> -	V	
c. Bare passive:	subj/th	Aux*	ag { * <i>meN</i> -/* <i>N</i> -/* <i>di</i> - }	V	
d. Object extraction:		Aux*	ag (* <i>me</i>) <i>N</i> -	V	____ _{th}

Recall that the subject from each of the clauses in (28a,b,c) can be A'-extracted, but no other nominal can. It is also possible to extract a theme across an agent without first passivizing, as in (28d), which we refer to as the “object extraction” clause type. (Only the gap left by object A'-extraction is indicated, not the moved object's landing site.) As noted above, object extraction shares with the bare passive the property of having an immediately preverbal agent, but differs in its use of the short active prefix *N*-. Object extraction also differs from the active clause type in the unavailability of the long active prefix *meN*- as well as its low agent position. In the following section, we present our analysis for these four clause types and the relationships between them.

We contrast these Desa clause types to the well-studied paradigm of clause types in SI and SM, summarized in (29).

(29) **Summary of SI/SM clause types:**

a. Active:	subj/ag	Aux*	<i>meN</i> -	V	obj/th
b. <i>Di</i> - passive:	subj/th	Aux*	<i>di</i> -	V	
c. Bare passive:	subj/th	Aux*	ag { * <i>meN</i> -/* <i>di</i> - }	V	
d. Object extraction:	subj/ag	Aux*	(* <i>meN</i> -)	V	____ _{th}

Comparing the SI/SM paradigm in (29) to the Desa pattern in (28), we notice two salient differences. First is that the nasal prefix in SI/SM does not have a short form, and therefore either appears as the long *meN*- prefix in (29a) or is fully absent in (29d), in contrast to the corresponding patterns in Desa. Second is that in SI/SM, but not in Desa, a nominal always occupies the high subject position, modulo its later A'-extraction. As reviewed above, the shared property of a high agentive subject in both active clauses (29a) and object extraction (29d) has led authors to describe the object extraction clause type in SI/SM as a form of active clause but with “deletion” of the active voice prefix *meN*-. In our discussion below, inspired by the pattern observed in Desa, we pursue the idea that object extraction in SI/SM is best described as sharing some features with active clauses and bare passive clauses, but formally distinct from both.

3 Malayic clausal syntax and object extraction in Desa

We now turn to our proposal for Malayic clausal syntax with an emphasis on the interaction of verbal morphosyntax and A'-extraction, discussed above. We discuss the syntax of the verb phrase and the analysis of voice alternations in section 3.1, drawing especially on the previous proposals in Aldridge 2008b and Nomoto 2015, 2021. Informed by the variation in verbal forms in Desa, we treat *me-* and *N-* as two separate prefixes on active verbs, which correspond to two distinct functional heads hypothesized by Nomoto (2015, 2021). We adopt the view that the verbal extended projection constitutes a cyclic domain (the *verbal phase* in Minimalist syntax), with movement necessarily proceeding through its “edge”. In section 3.2, we show how this assumption, together with the detailed organization of the verbal domain that we propose, derives the basic subject-only restriction on nominal A'-movement. We then discuss the syntax of object extraction clauses in Desa in section 3.3.

The syntactic proposal that we develop here for Desa in section 3.1 serves as a template for the analysis of similar morphosyntactic patterns in related languages of the region. In section 4, we extend our analysis to the standard languages. Clarifying the nature of phasehood and exactly how phases enforce derivational cyclicity, we will arrive at an explanatory new understanding of the well-known “*meN*-deletion” pattern of object extraction clauses in SI and SM, and show how our analysis accounts for many details of their behavior. In section 5, we show how our proposal can address the range of attested variation in verbal morphology across Malayic and related languages and dialects, and in particular how our proposal accurately predicts further correlations between the morphological forms of verbs and movement possibilities.

3.1 Malayic clausal syntax and voice alternations

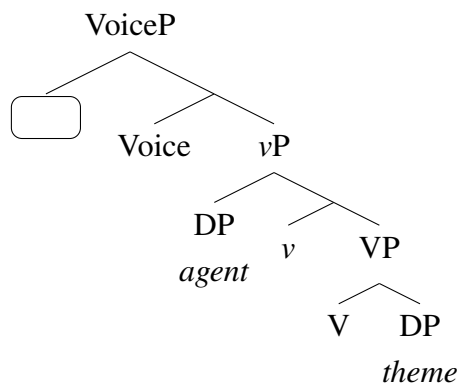
We begin with our proposal for the basic clausal syntax of Malayic languages. Our proposal here synthesizes the key insights of previous proposals for SI and SM syntax, in particular closely following the proposals for the formal licensing of nominal arguments in Aldridge 2008b, the organization of the verb phrase in Nomoto 2015, 2021, and the idea that the verbal domain constitutes a cyclic domain with limited permeability for extraction. In this section, we will specifically illustrate its application to the verbal morphosyntax of Desa, being careful to flag Desa-specific points and novel components of our account. We then discuss the application of this proposal to SI and SM and other related languages in sections 4 and 5.

Many aspects of our proposal here reflect widely adopted ideas in contemporary Minimalist syntax, especially for the analysis of languages with dominant S(Aux)VO word order, which will be familiar to many readers and which we treat here as assumptions of our framework of analysis. We begin with three such

ideas — on the base positions of arguments, the nature of the subject position, and the need of nominals to be licensed — and then move to the more Malayic specific components of our analysis, again building on prior proposals for SI and SM.

Thematic positions in VoiceP We assume that all arguments in a clause are generated within the extended projection of the verb, in designated positions associated with their thematic roles (Baker, 1988, a.o.), including agentive subjects (the so-called *predicate-internal subject hypothesis*; see Kitagawa 1986, Kuroda 1988, and Kratzer 1996, as well as Guilfoyle et al. 1992 for SI and SM). Themes are generated as complements of the lexical verb, V, and agents (or more generally, *external arguments*) are generated as specifiers of a functional head *v*, as in (30).

(30) **The base positions of arguments in VoiceP:**



We follow Nomoto 2015, 2021 in the view that there is an additional functional head, *Voice*, which takes *vP* as its complement.¹⁰ We will discuss *Voice* and its specifier position (boxed in (30)), which will play an important role in our analysis, below.

The subject position We derive the S(Aux)VO base order of the language as reflecting a high, dedicated subject position that is filled by moving a nominal argument out of the predicate domain. We assume that all clauses project a head T above VoiceP and other functional projections projecting auxiliaries, if any. T has a requirement that a nominal move to fill its specifier, Spec,TP, resulting in the subject linearly preceding any and all auxiliaries. This requirement is frequently called an “EPP requirement” (originally from Chomsky, 1982: 10; see also Chomsky 1995: 49, 213, Svenonius 2002, Joutteau, 2007: 182ff, a.o.) and we adopt this term here.

¹⁰ Such an organization is briefly suggested for SI by Cole, Hermon, and Yanti (2008: 1532–1533) and Kaufman (2014: 115–117). It is also assumed and illustrated in passing by Nomoto (2016: 187, 2020: 154). See also note 13 below.

Nominal licensing We adopt the view that all noun phrases require grammatical *licensing* — also known as *abstract Case* (Vergnaud, 1977; Chomsky, 1980, 1981; see also Sheehan, 2018 and citations there) — and that the limited inventory of sources of licensing plays a key role in constraining the space of possible grammatical derivations.

We adopt the proposal for mechanisms of nominal licensing in Indonesian-type languages developed in Aldridge 2008b and as adapted in Nomoto 2015, 2021, where there are two varieties of the head v which differ in their licensing abilities: v_{ACT} licenses one nominal that it c-commands and v_{PASS} licenses the nominal in its specifier, if one is projected. Both bare passives and *di-* passives involve the same v head, v_{PASS} , which can optionally introduce an external argument as its specifier. The morpheme *di-* will appear when no argument is projected in the specifier of v_{PASS} , as we detail below.¹¹ In addition, T licenses the nominal that it attracts to Spec,TP. In clauses with additional arguments such as ditransitives, we assume an additional functional head (e.g. Appl) serves to license an additional nominal.

Aldridge (2008b) refers to these three modes of nominal licensing — of the subject by T, of an internal argument by v_{ACT} , and of Spec, v P by v_{PASS} — as assigning “nominative”, “accusative”, and “ergative” case, respectively, and Nomoto (2021: 64) follows this terminology. This terminology reflects the fact that these licensing configurations reflect what has been proposed as influential generative accounts of morphological nominative, accusative, and ergative case, respectively (see e.g. Aldridge 2004, 2008a, Legate 2008, 2012b, Woolford 2006, and citations there). In other words, for Aldridge, active clauses exhibit accusative abstract Case alignment whereas bare passives exhibit ergative alignment. However, as nominals in these languages by and large do not express morphological case distinctions (although see note 18), we choose to describe this aspect of our proposal as reflecting a formal requirement of nominals to be “licensed”, rather than a requirement for “(morphological) case”.

Considerations of nominal licensing govern the relationship between the choice of v head and the choice of subject, as we explicate below. The basic logic is as follows for a simple bivalent verb. If the derivation involves v_{ACT} , v will license the theme, but the agent has no source of licensing within VoiceP. The agent must therefore be moved to Spec,TP and licensed by T. If the derivation instead involves v_{PASS} , v will license the agent in Spec, v P if projected, but the theme will lack licensing, so it must be moved to Spec,TP and

¹¹ Our v_{PASS} corresponds to two different heads in Aldridge’s (2008b) proposal: what she calls v_{Erg} which hosts and licenses an external argument as its specifier and v_{Pass} which does neither. Our unified approach here echoes an analytic intuition expressed by Nomoto (2021: 64), who views the lack of licensing for an internal argument within VoiceP to be the shared and defining property of *di-* passives and bare passives as both “passives”.

In SI/SM as well as in Desa, agents of *di-* passives can be introduced as postverbal adjuncts, as a DP or PP. If a DP is adjoined, it would need a source of licensing which might be provided by adjacency to the verb (Levin, 2015; Erlewine, 2018; Erlewine et al., 2020).

licensed by T. In this way, the choice of v head correlates indirectly with the choice of nominal that moves to Spec,TP to become the subject, due to the need for each noun phrase to be licensed. We present each of these grammatical and ungrammatical derivational possibilities in greater detail below.

The morphology of Voice and v We follow Nomoto 2015, 2021 quite closely in the organization of the verbal domain, with heads Voice and v (see the tree in (30) above), and their functions in active and passive clauses. In this subsection, we discuss the morphological forms associated with these functional heads. We will show how active and passive verbal forms correlate with different choices of subject later in this section.

As introduced above, active clauses involve v_{ACT} , whereas *di*- passives and bare passives involve v_{PASS} . In addition, Nomoto proposes the higher functional head Voice as the locus of voice morphology — for Nomoto, *meN*- or *di*-/Ø — whose realization correlates with the choice of v head. Nomoto enforces the correlation between the choice of v head and the choice of voice prefix pronounced at Voice “by means of a selectional restriction” (Nomoto, 2021: 64), but we will instead pursue the view that there is just one variety of Voice head in the syntax, whose realization varies as a matter of allomorphic variation.

Adopting Nomoto’s proposal for the verbal extended projection with two functional heads, Voice and v , facilitates the analysis of voice prefixes in Desa and is in turn supported by it. Recall that in Desa, active verbs can be introduced by *meN*- or *N*-, and that these two forms are in free variation in active clauses, but importantly behave differently in the context of object extraction. We propose that in the Desa active voice, v realizes the homorganic nasal *N*- and Voice is optionally realized as the syllabic prefix *me*-. For *di*- passive clauses, we maintain Nomoto’s position that Voice realizes *di*-. We sketch the structure of the VoiceP in the three basic clause types and their morphological realization in (31) below:

(31) **Desa voice morphology in Voice and v :**

	[VoiceP	Voice	[v P	v +V	...
a. Active:		(<i>me</i> -)		<i>N</i> -V	
b. <i>Di</i> - passive:		<i>di</i> -		V	
c. Bare passive:			DP _{ag}	V	

We argue in section 5.1 that this morphological decomposition of *meN*- across two syntactic heads is in fact motivated for SI and SM.

We adopt *Distributed Morphology*, a realizational model of morphology where the surface forms of terminals are determined from the syntactic structure by the application of appropriate pronunciation rules, called *vocabulary items*. (See overviews in Harley and Noyer 1999, Embick and Noyer 2007, and Bobaljik 2017, and citations there.) We give the vocabulary items for the heads Voice and v in Desa in (32), where

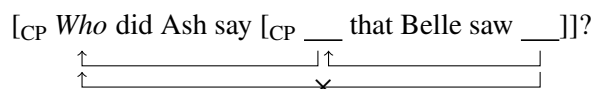
* indicates linear adjacency (Marantz, 1988; Embick, 2010; a.o.). We treat the optional realization of *me-* formally as the optional application of the vocabulary item in (32ai). When Voice is realized overtly, as *me-* or *di-*, it affixes to the verb in *v* via *Local Dislocation* (Embick and Noyer, 2001), a postsyntactic operation that requires linear adjacency between Voice and *v* to apply.

(32) **Vocabulary items in Desa:**

- | | |
|---|--|
| a. i. (Voice $\leftrightarrow$ <i>me-</i> / ____ * v_{ACT}) | b. $v_{\text{ACT}} \leftrightarrow N-$ |
| ii. Voice $\leftrightarrow$ <i>di-</i> / ____ * v_{PASS} | c. $v_{\text{PASS}} \leftrightarrow \emptyset$ |
| iii. Voice $\leftrightarrow \emptyset$ / elsewhere | |

VoiceP as a phase A long-standing intuition in generative syntax is that complex syntactic structures are built up cyclically, in chunks, prohibiting the formation of syntactic dependencies in a truly unbounded fashion. For instance, even apparent cases of long-distance (especially cross-clausal) movement involves multiple, more local (i.e. *successive-cyclic*) steps of movement. In the English example (33), long-distance *wh*-movement has been argued to first stop at the intermediate Spec,CP position and then move to its surface position (see e.g. Chomsky 1973, 1977 and a recent overview in Van Urk 2020).

(33) **Long-distance movement is successive-cyclic:**



In contemporary Minimalist syntax, the relevant cyclic units with limited permeability for extraction are called *phases*. Chomsky (2000, 2001) proposed that phases are subject to the *Phase Impenetrability Condition* in (34):

(34) **Phase Impenetrability Condition (PIC):** (Chomsky 2000: 108, 2001: 13)

In phase α with head H, the domain [complement] of H is not accessible to operations outside α , only H and its edge [specifiers] are accessible to such operations.

Assuming that full clauses (CPs) are phases, movement directly from the embedded object position to the surface position in (33) would violate the PIC. The *wh*-phrase must first move to the embedded clause's “edge”, from which it can later move out of the phase. For our discussion in this section, it suffices to simply assume Chomsky's PIC in (34), but later in section 4, we will advocate for an alternative and more principled conception of the cyclicity effects of phases as derived from the role of phases in word order determination (as in Fox and Pesetsky, 2005).

In addition to full clauses (CPs) and full noun phrases (DPs), many works in contemporary Minimalist syntax make reference to the idea that the full verbal extended projection is a phase. Adopting this idea of a clause-medial *verbal phase*, we propose that the VoiceP projection in Malayic constitutes a phase. The *v*P projection does not.¹²

Many previous Minimalist accounts for voice and extraction in SI and SM also make use of the idea of a verbal phase, including Aldridge 2008b, Cole et al. 2008, Sato 2012, Jeoung 2017, 2018a,b, and Legate 2014. However, all of these works make reference to a single head (called Voice or *v*, depending on the work) which both introduces the external argument as its specifier and serves as the phase head. Our proposal here that VoiceP is a phase can be thought of as the simple synthesis of this idea of a verbal phase with Nomoto's organization for the extended verb phrase with the two separate heads, Voice and *v*. Separating the functions of serving as the phase head and introducing the external argument across two different functional heads, as Nomoto's phrase structure allows us to do, will play a crucial role in our analysis.¹³

Specifiers of VoiceP In addition to being a phase, we propose that the Malayic VoiceP has a particular organization of specifiers, in (35) below:

(35) **Specifiers of the Malayic VoiceP:**

- a. VoiceP always has exactly one nominal (DP) specifier (boxed in (30) above); and
- b. VoiceP may optionally host a non-nominal specifier.

The crucial distinction between nominal versus non-nominal specifiers will be motivated by PP extraction facts as in (40–41) below, and echoes widespread differences in the behavior of nominal versus non-nominal A'-extraction constructions in many Austronesian languages (see e.g. Erlewine, 2018; Hsieh, 2020: ch. 7). We also note that functional heads with precisely this organization — having exactly one nominal specifier, with no restriction on non-nominal specifiers — have also been proposed for Dinka (Van Urk and Richards, 2015: 129–133) and Kipsigis (Bossi and Diercks, 2019: 16–19), two Nilotic languages.

The requirement for a nominal specifier (35a) is satisfied by movement of an accessible nominal from within VoiceP. The choice of nominal to be moved will be determined by considerations of nominal licensing:

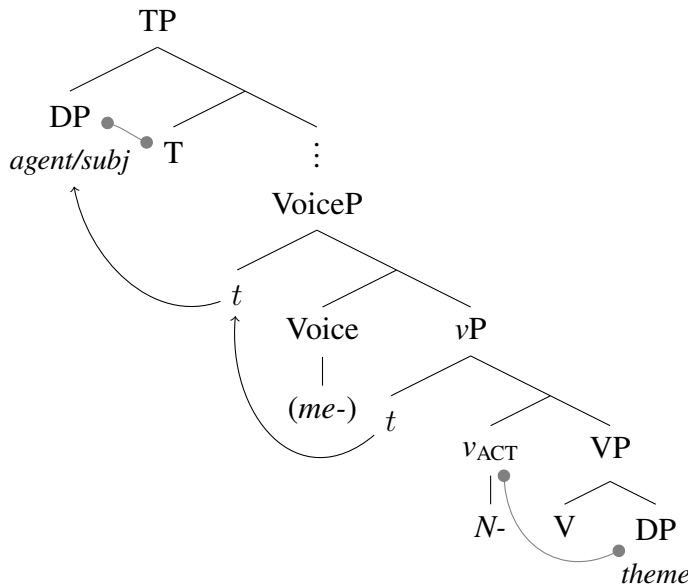
¹² Nomoto (2015, 2021) himself does not discuss the verbal domain as a phase, which reflects the fact that accounting for restrictions on A'-extraction is not a goal of these works. Our proposal here that Voice rather than *v* is a phase head accords with the conjecture as in Bošković 2014 that the highest head of an extended projection constitutes the phase head.

¹³ The overall geometry of the verbal phase here — taking the proposal for the extended projections of the verb in Nomoto 2015, 2021 together with the idea that VoiceP constitutes a phase — parallels structures that have been proposed for some other languages: see especially Gallego 2008, Richards 2010, 2023, Coon et al. 2014, Brodtkin 2022, 2025, Pietraszko 2023, Hsieh 2025, as well as Collins 2005 on passives and a brief suggestion in Cole et al. 2008: 1532–1533. These proposals all involve a higher phase head, Voice, with a *v*P complement, where *v* introduces the external argument.

the nominal that moves to Spec, VoiceP must be the one that lacks licensing within VoiceP, as it must move to Spec, TP to become the subject of the clause, as we illustrate below.¹⁴ In active clauses, this will be the agent. In passive clauses, it will instead be another nominal argument, such as the theme, that moves to Spec, VoiceP.

Derivations We now illustrate the structure of full clauses in each of the voices, beginning with an active clause. The v will be the active variety, v_{ACT} . The v_{ACT} head licenses the theme, which it c-commands. The agent moves to become the one nominal specifier of VoiceP. T subsequently attracts the agent from Spec, VoiceP to satisfy its own need to have a nominal specifier (the EPP requirement) and licenses it. Licensing relationships are indicated by $\bullet\bullet$ in gray below. We propose that this derivation underlies the “DP_{agent} Aux* Voice+v+V DP_{theme}” word order of active clauses in Malayic languages. We illustrate this structure in (36), with morphological realizations for Voice and v according to the Desa vocabulary items in (32) above. (Head movement of V to v is not shown.) This is also our proposal for the structure of active clauses in other Malayic languages, but with minor differences in the morphological realization of Voice and v , as we discuss in sections 4 and 5.

(36) **Malayic active TP, with Desa morphology:**

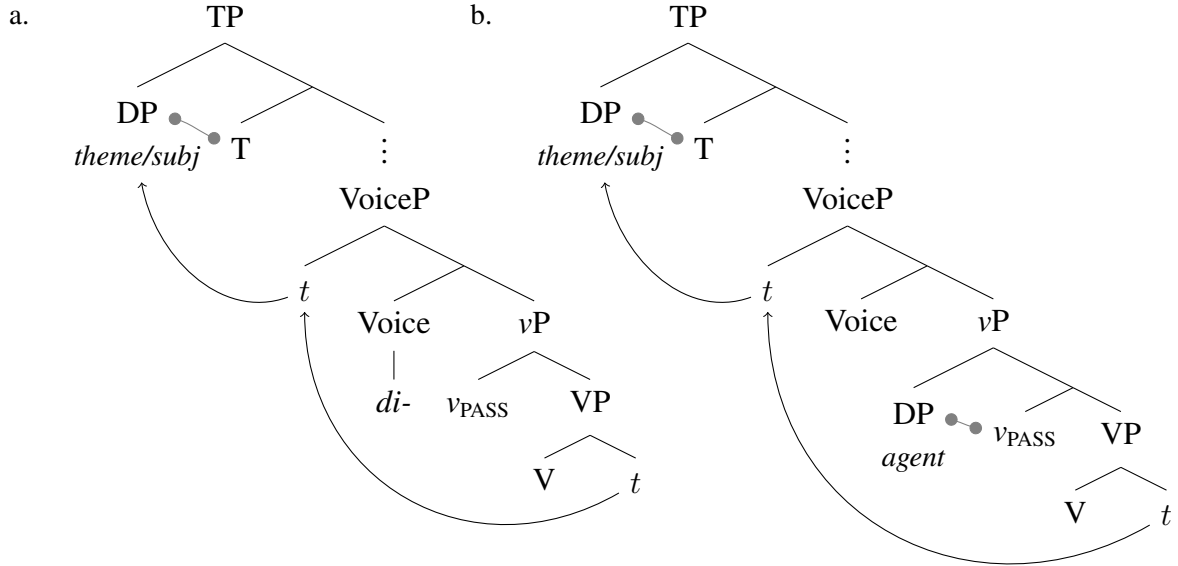


Next, we turn to the structure of passive clauses. We present two versions of this derivation: one where v_{PASS} does not project a specifier in (37a), and one where it does in (37b). In the latter case, the v_{PASS} head itself licenses the specifier. The two derivations are otherwise equivalent: An internal argument (here, the

¹⁴ For the purposes of our presentation here, we assume that movement may apply freely to satisfy the specifier requirement, in the style of Chomsky’s (1981) “Move-alpha”, sometimes described as “Move anything anywhere” (see e.g. Saito, 1985: 42). Derivations that fail to satisfy the requirement for nominal licensing or the specifier requirements on VoiceP (35) will then be filtered out.

theme) cannot be licensed within VoiceP. This nominal moves to become the one specifier of VoiceP, and then moves to become the subject in Spec,TP and is licensed by T. These derivations in (37) yield passive clauses with the word order “DP_{theme} Aux* Voice (DP_{agent}) v+V”.

(37) **Malayic passive TPs:**



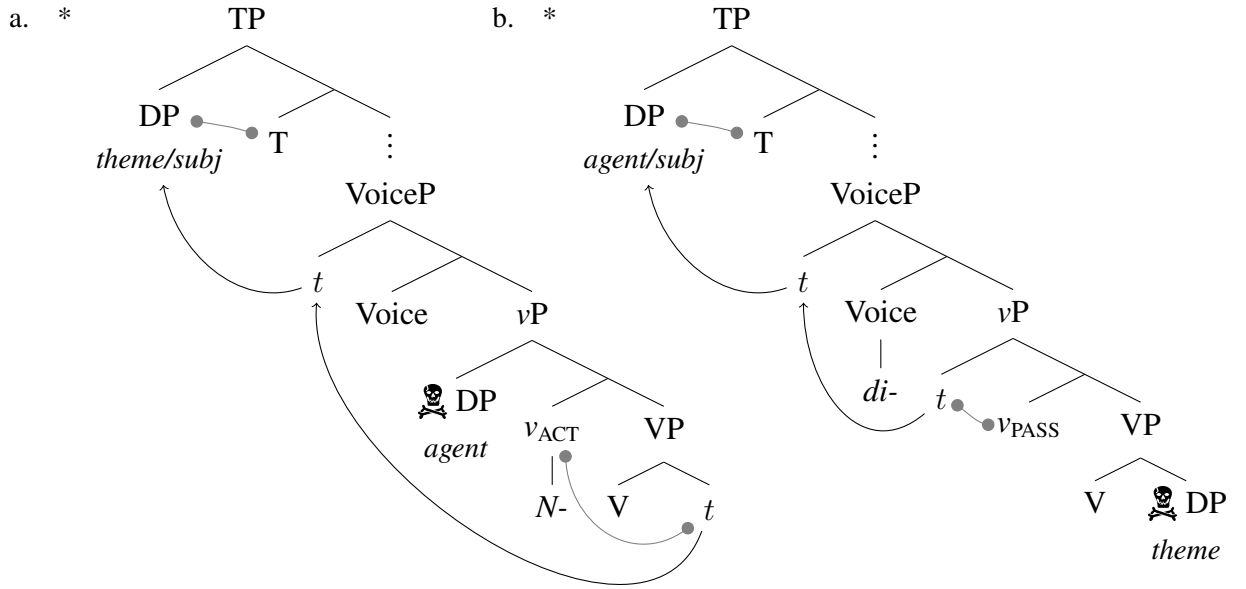
As per our vocabulary items in (32) above, Voice will realize *di-* in (37a) as it is linearly adjacent to v_{PASS} ; in contrast, Voice will be unpronounced in (37b) as the agent linearly intervenes between Voice and v_{PASS} , resulting in a bare passive. Keeping the assumption fixed that Voice, when overt, must prefix to the verb in v via Local Dislocation, which requires linear adjacency, the organization of the verbal extended projection here adopted from Nomoto 2015, 2021 together with our vocabulary item for Voice as *di-* in (32aii) above offers a natural explanation for the complementary distribution of voice prefixes and immediately preverbal non-subject agents, which is widely observed across a wide range of Indonesian-type languages (see e.g. Ross 2002, Donohue 2008, Nomoto 2021).¹⁵

Mismatched derivations For completeness, we illustrate hypothetical derivations where a different nominal argument moves via Spec,VoiceP to the subject position, Spec,TP. The two such logical possibilities with a simple bivalent verb are illustrated in (38) below. (38a) is a derivation with v_{ACT} and a theme subject, whereas (38b) involves v_{PASS} with an agent subject. Based on our proposal for Desa verbal morphology above in (32), if grammatical, these derivations would yield sentences of the form “DP_{theme} Aux* DP_{agent} N-V” in (38a) and

¹⁵ The Kendayan languages of western Borneo constitute the one counterexample to this generalization amongst Malayic languages, as highlighted by Adelaar (2005b,c), Kaufman (2007), and Sommerlot (2024). See section 5.2 and examples in (74) there.

“DP_{agent} Aux* *di*-V DP_{theme}” in (38b).

(38) **Malayic TP derivations with the wrong nominal chosen as subject:**



In both of these derivations, the nominal that already *has* licensing by *v* — the theme licensed by v_{ACT} in (38a) and the agent licensed by v_{PASS} in (38b) — moves to Spec,VoiceP and then to the Spec,TP subject position to satisfy T’s EPP requirement. T licenses the nominal that it moves to Spec,TP, so these subject nominals then may be licensed redundantly by T; it is not possible for T to instead license the other, non-subject nominal below. Both of these structures will then be ungrammatical, even though there are no problems with the sequence of derivational steps involved, as they will leave a nominal with no source of licensing, indicated by the skull (as violations of the Case Filter, in abstract Case terms).

Summary To conclude this section, we schematically represent the full TP structures that we propose for active, *di*- passive, and bare passive clauses in Malayic languages in (39) below. Here we indicate the options for the realization of voice prefixes in Desa, but we propose that the same syntactic structures apply for these clause types in other Malayic languages as well, as we discuss below.

(39) **TP clause structures for the three voice types, with Desa morphology:**

	[TP	...	[VoiceP	Voice	[_v P	v+V	
a. Active:	DP _{ag}	Aux*	<i>t</i>	(<i>me-</i>)	<i>t</i>	N-V	DP _{th}
	↑		↑		↑		
b. <i>Di-</i> passive:	DP _{th}	Aux*	<i>t</i>	<i>di-</i>		V	<i>t</i>
	↑		↑		↑		
c. Bare passive:	DP _{th}	Aux*	<i>t</i>		DP _{ag}	V	<i>t</i>
	↑		↑		↑		

The traces in these structures reflect the movement paths of the subject. In each case, the subject first moves from its base position to Spec,VoiceP, to satisfy the requirement that VoiceP host exactly one nominal specifier. It then moves from this position to the high subject position, Spec,TP. The intermediate movement step through Spec,VoiceP is necessary because VoiceP is a phase; in Chomsky's terms, movement from the complement of the phase head (from inside *v*P) directly out of the phase to Spec,TP would violate phase impenetrability (see 34).

3.2 Restrictions on A'-extraction

Next we turn to the possibilities for A'-extraction, by which we refer to processes of relativization, *wh*-movement, focus fronting, and topicalization. We assume that all A'-extractions reflect movement triggered by the complementizer C, which takes TP as its complement. All movement steps must satisfy the locality restrictions imposed by phase impenetrability.

We first consider the A'-extraction of nominals. If we start with any of the TP structures that we proposed in the previous section, the only nominal that can be attracted by C for A'-movement is the subject. The subject is in Spec,TP, outside of the VoiceP phase, and moved there by passing through Spec,VoiceP. As we proposed above, VoiceP always hosts exactly one nominal specifier, making it impossible to move any other nominal out of the VoiceP phase. Our proposal above therefore immediately predicts the basic subject-only restriction on the A'-extraction of nominals in these languages. We will later discuss a distinct derivation that underlies object extraction clauses, for Desa in section 3.3 and for other Malayic languages including SI and SM in section 4.2.

Our proposal also correctly predicts that non-nominal constituents can be A'-extracted from within VoiceP, without any restrictions on voice choice. See for example the fronting of a *wh*-containing PP for the Desa *where*-questions below, from an active clause (40a) or from a *di-* passive clause (40b). The examples in (41) show the grammaticality of PP topicalization and *wh*-fronting in SI from an active (41a), *di-* passive active (41b), and bare passive clause (41c). (We predict PP extraction from Desa bare passive clauses to be

grammatical as well, but this was not tested.) Note in particular that the A'-extraction of PPs from an active clause does not require a prefix-less verb, unlike the extraction of non-subject nominals.

(40) **PP A'-movement is not sensitive to voice, in Desa:**

- a. [PP Di oni] inya { m-oli / mem-oli } beju to ____?
 LOC where 3sg N-buy MEN-buy shirt this
 'Where did s/he buy this shirt?'
- b. [PP Di oni] beju to di-beli oleh inya ____?
 LOC where shirt this PASS-buy by 3sg
 'Where was this shirt bought by him/her?'

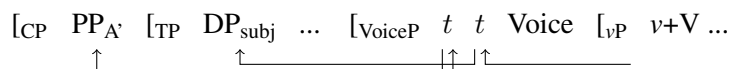
(41) **PP A'-movement is not sensitive to voice, in SI/SM:**

(SI; (a): Chung, 1976a: 78; (b–c): Chung, 1976b: 82)

- a. [PP Kepada Ani], Hasan meng-irim-kan surat itu _____. (*meN-kirim-kan* > *mengirimkan*)
 to Ani Hasan ACT-send-APPL letter that
 'To Ani, Hasan sent the letter.'
- b. [PP Kepada siapa] bunga itu di-beri-kan ____?
 to who flower that PASS-give-APPL
 'To whom were those flowers given?'
- c. [PP Kepada siapa] bunga itu kau beri-kan ____?
 to who flower that 2sg give-APPL
 'To whom did you give those flowers?'

As noted in (35b) above, we hypothesize that the Malayic VoiceP can host non-nominal specifiers, in addition to its one nominal specifier.¹⁶ A non-nominal constituent originating within VoiceP can therefore move first to an additional specifier of VoiceP, feeding A'-movement to the clause periphery. Such a derivation is schematized in (42) below, abstracting over the choice of voice and therefore the VoiceP-internal base position for the subject.

(42) **Non-nominal A'-extraction fed by additional specifier to VoiceP:**



¹⁶ Our editor John Beavers asks whether examples with apparent extraction of non-nominals out of VoiceP, as in (40–41), could instead be derived by base-generating the phrase outside of VoiceP. Such an alternative parse is unlikely for (41), where the PP is an argument of the verb. In addition, we note that non-nominals can also be A'-extracted from embedded clauses, again across voice-prefixed verbs, as in example (65) below.

3.3 Object extraction in Desa

Our proposal thus ensures that the subject is the only nominal that can be A'-extracted from Malayic clauses built in the manner described in section 3.1 above. However, as we saw in section 2.3 above, Desa allows for another type of clause where an internal argument is A'-extracted, apparently without first undergoing passivization via a *di-* passive or bare passive. We refer to such structures as “object extraction clauses” and give another such example in (43). In this section, we show how we can derive Desa object extraction clauses with minimal adjustment to our overall proposal.

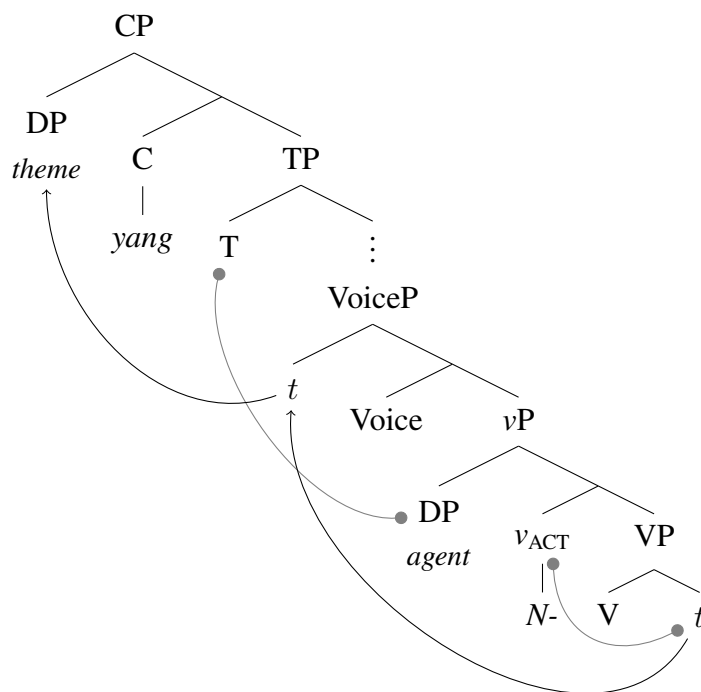
(43) **An example of Desa object extraction:**

Opai yang tengah inya { **m-ilau** / ***mem-ilau** } ____? (*N-pilau* > *milau*)
what C PROG 3sg N-look.for MEN-look.for
'What is s/he looking for?'

As noted above, Desa object extraction clauses as in (43) defy straightforward classification in terms of the three basic voices that we have discussed. First, the verb in object extractions bears the short active prefix *N-*, and the long active *meN-* is ungrammatical. This is unlike active clauses where the two active forms are in free variation, and is also unlike the passive clauses where the verb must bear a *di-* prefix or no prefix at all. Second, the agent is in immediately preverbal position, following any auxiliaries such as *tengah* in (43). The agent cannot precede auxiliaries, as we showed in example (26) above. The position of the agent thus makes object extraction clauses seem most similar to the bare passive, but we note again that bare passive clauses disallow any voice prefix, including the short prefix *N-*; see (12).

Motivated by the above properties, we propose to think of Desa object extraction clauses in the following way, as illustrated in (44) below. Object extraction clauses have a VoiceP structure with the head v_{ACT} , which licenses an internal argument (here, the theme) and realizes the short active prefix *N-*. (We discuss the unavailability of long *meN-* below.) However, unlike canonical active clauses involving v_{ACT} , the internal argument then moves to become the sole nominal specifier of VoiceP, which later moves to Spec,CP. The agent stays in its Spec, v P base position, explaining its immediately preverbal position. We discuss the agent's source of licensing below.

(44) **Deriving Desa object extraction:**



A notable feature of this structure in (44) is that no phrase occupies the conventional, high subject position of Spec,TP. This is in contrast to the regular clause types, where Spec,TP is always filled by the nominal that T licenses (the so-called EPP requirement). Work on the interaction of A'-extraction and subjecthood has shown that, in some languages, the EPP requirement is relaxed specifically when a nominal is A'-moved to Spec,CP.¹⁷ We hypothesize that Desa is one such language: in clauses involving nominal A'-extraction, T licenses a nominal that it c-commands, without moving it to Spec,TP. This is the source of licensing for the agent in Spec,vP.¹⁸ Note that we will argue that object extraction clauses in SI/SM differ from Desa on this point, as we discuss in section 4.2 below.

The agent staying in its Spec,vP base position serves to explain the verbal morphology of Desa object extraction clauses. Recall that the verb in such clauses bears the short active prefix *N-*, unlike in bare passives,

¹⁷ Motivating examples include non-subject A'-constructions with postverbal subjects in Romance (see e.g. Barbosa, 2001 and citations there) and Kilega (Bantu; Kinyalolo, 1991; Carstens, 2005), Scandinavian stylistic fronting (see e.g. Holmberg, 2000; Sigurðsson, 2010) and English *as*-parentheticals (Postal, 2004) and locative inversion (Bruening, 2010). See for example Carstens 2005, Legate 2011, 2014, and Erlewine 2018, 2020 for formal proposals for such interactions.

¹⁸ We assume that licensing (via Agree) is unaffected by intervening phase boundaries, unlike movement, as per Bošković 2003, 2007. This assumption accords with the Cyclic Linearization view of phase impenetrability, which we adopt and advocate for, in section 4.2 below.

A reviewer asks if there might be evidence that the low agent in Desa object extraction is licensed by *v*, rather than T. Recall that immediately preverbal agents of bare passives allow for reduced, proclitic forms for agent pronouns; Nomoto (2021) proposes that this reflects a case distinction, in that bare passive agents are assigned inherent ergative case from *v*_{PASS}, following Aldridge 2008b. We would therefore like to know whether Desa object extraction clauses also allow for such proclitic forms for their immediately preverbal agents. Unfortunately, we cannot answer this question based on our existing data.

but cannot bear the long *meN-* active prefix, unlike in active clauses. See example (43). This is precisely what we predict from the derivation for object extraction clauses in (44) as well as our proposal for the allomorphic realization of Voice and *v* in Desa. We repeat the vocabulary items for Desa from (32) for the verbal functional heads in object extraction clauses as in (44), in (45):

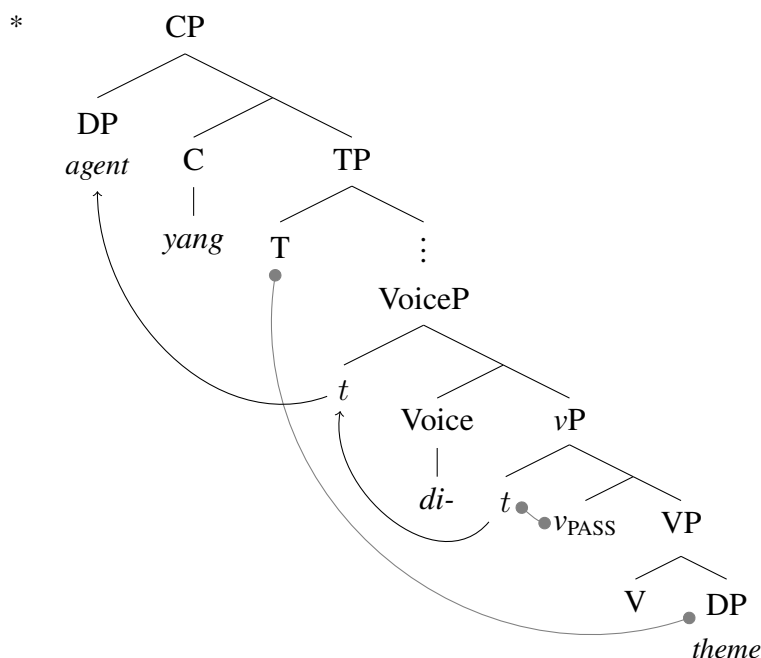
- (45) **Vocabulary items for the functional heads in Desa object extraction clauses:** from (32)
- a. $v_{\text{ACT}} \leftrightarrow N-$
 - b. $(\text{Voice} \leftrightarrow me- / __ * v_{\text{ACT}})$
 - c. $\text{Voice} \leftrightarrow \emptyset / \text{elsewhere}$

First, as also noted above, the prefix *N-* appears as the realization of v_{ACT} , which is necessary to license the object that then A'-moves to Spec,CP in (44). Second, *me-* cannot appear because the agent linearly intervenes between Voice and *v*, therefore bleeding the application of the vocabulary item in (45b). In addition, even if Voice were to be realized as *me-* in this structure, it would then fail to affix to the verb, as we discuss further in section 4.3.

In summary, the specification that Voice's overt allomorphs are prefixes, together with the low position of agents in Spec,*v*P in object extraction clauses (as well as in bare passives), conspires to require that Voice be realized by a null allomorph in this configuration. In contrast, the presence of the low agent does not affect the pronunciation of *v* as *N-*, which appears transparently in Desa due to the possibility of pronouncing *N-* without *me-* in the language, unlike in SI and SM.

Finally, for completeness, given the possibility of moving the theme to Spec,VoiceP in a derivation with v_{ACT} as in (44) above, we briefly discuss the potential possibility of moving the agent to Spec,VoiceP in a derivation with v_{PASS} . Consider the structure in (46) below, which we claim is ungrammatical.

(46) **Hypothetical agent A'-extraction with v_{PASS} :**



The basic logic of (46) is as follows. We start with a derivation with v_{PASS} , which licenses the agent, and then move that agent to Spec, VoiceP. The theme has no source of licensing within VoiceP. The agent A'-moves to Spec, CP, leaving Spec, TP empty as per our discussion above (see note 17). T licenses the in-situ theme, which it c-commands (see note 18), without moving it. Therefore, based on our general analysis for Malayic clause structure and the Desa-specific amendment in this section, the derivation in (46) would be predicted to be grammatical.

Based on our vocabulary items for Desa in (32) above, we would predict Voice in (46) to realize *di-* as it is linearly adjacent to v_{PASS} . Thus, if the derivation in (46) were indeed possible, we would predict the availability of an agent A'-extraction clause with a *di-* passive verb and in-situ internal argument. This is however not possible, as reflected by the ungrammaticality of the *di-* form variant of the agent relative clause in (47) below. Agent A'-extraction requires an active clause, which in Desa may involve a short or long active prefix.

(47) **Agent A'-extraction in Desa is ungrammatical with *di-* verb:**

Aku nge-liet lelaki [_{RC} yang { ng-ikat / meng-ikat / *di-ikat } tali yetn].
 1sg N-see man C N-tie ME_N-tie PASS-tie rope DEM
 'I saw the man that tied the rope.'

We tentatively suggest that the ungrammaticality of (47) with *di-* (and more generally, structures of the form in (46)) reflects a typologically familiar restriction on movement. Recall that Aldridge (2008b) and

Nomoto (2021), who we follow in our proposal for nominal licensing, describe licensing of the specifier of νP by ν_{PASS} as an instance of abstract ergative case assignment. Many (though not all) morphologically ergative languages exhibit a ban on the A' -movement of ergative arguments; see e.g. Dixon 1994, Manning 1996, and Aldridge 2008a. Although the Malayic languages including Desa do not exhibit robust nominal case distinctions, the ungrammaticality of the agent A' -extraction derivation sketched in (46) above appears to reflect this widely attested ergative extraction restriction. See Deal 2016 and Polinsky 2016, 2017 for two overviews of analytic approaches to such extraction restrictions, as well as Drummond 2022, 2023 for recent arguments for the possibility of such syntactic ergativity without overt morphological case distinctions.

Here we leave open the question of the exact nature of the ungrammaticality of (46) and the nature of ergative extraction restrictions more generally, but we mention one concrete possibility. Erlewine (2016a, 2017) proposes that A' -movement is subject to an “anti-locality” constraint that bans A' -movement from the specifier of a projection XP to the specifier of the projection that immediately dominates XP , but A -movement (i.e. movement (on the way) to a licensing position) is not subject to this constraint.¹⁹ Notice that movement of the agent from Spec, νP to Spec,VoiceP in (46) is from its licensing position, and thus A' -movement, in violation of Erlewine’s anti-locality constraint. In contrast, movement of the agent from Spec, νP to Spec,VoiceP in the grammatical derivation (36) above is A -movement, as the agent is not licensed within VoiceP.

4 Explaining “*meN*-deletion” in Standard Indonesian and Malay

We now extend our analysis to Standard Indonesian and Malay (SI/SM), where the interaction of voice and A' -extraction has received significant attention in prior theoretical literature. We begin in section 4.1 by briefly presenting our proposal for the basic clause types in SI and SM, which follows our proposal for Desa above, with only a minor difference in morphological realization. Then in section 4.2, we discuss the syntax of object extraction in SI and SM, which differs from that of Desa in that both the A' -moving nominal and the subject simultaneously move out of the verbal phase. We will show that grammatical patterns of object extraction, including the obligatory disappearance of the *meN*- prefix in such clauses (“*meN*-deletion”), can be derived from our overall proposal together with Fox and Pesetsky’s (2005) *Cyclic Linearization* theory for the nature of phase impenetrability effects. We compare our account for the “*meN*-deletion” effect to preceding accounts in section 4.3, followed by discussion of further positive predictions and evidence for our account.

¹⁹ See also Bošković 2016, Deal 2019, Erlewine 2020, Branen 2023 for further discussion and motivation for this variety of anti-locality constraint, but described as applying to both A - and A' -movements.

4.1 Clause structure and voice realization in SI/SM

As noted above, we intend for our proposal for the basic clausal syntax of Desa (in section 3.1 above) to extend to other Malayic languages including SI and SM, as well as to related languages of the region that have also been described as having so-called Indonesian-type syntax (see e.g. Wolff, 1996; Ross, 2002). Here we reiterate the key features of our proposal for the basic clause types and then discuss how we capture the one point of morphological divergence between Desa and SI/SM below. We discuss the further extension of our proposal to other languages of the region in section 5 below.

The most important feature of our proposal for Malayic syntax presented above is the organization of the verbal phase, VoiceP. Voice always hosts exactly one nominal specifier, which moves there from a lower thematic position. In the basic clause types, this one nominal is the subject. In an active clause, the subject is the external argument which originates in Spec, ν P; in a passive clause, it is an internal argument generated lower in the clause. The subject then moves up to a high subject position (Spec,TP), so it linearly precedes any auxiliaries projected above VoiceP.

We give schematic structures for the three basic clause types — active, *di*- passive, and bare passive — in (48) below, repeated from (39) above, with minor adjustment to the realization of active functional heads. The syntactic derivations for these three clause types in SI and SM are exactly the same as in Desa. We reiterate that this proposal adopts and synthesizes the strategies for nominal licensing from Aldridge 2008b and the organization of the verbal extended projection from Nomoto 2015, 2021. These mechanisms ensure that active clause derivations as in (48a) involve ν_{ACT} whereas the passive derivations as in (48b,c) involve ν_{PASS} . See detailed discussion in section 3.1 above.

(48) **TP clause structures for the three voice types:** based on (39)

	[TP	...	[VoiceP	Voice	[ν P	ν +V	
a. Active:	DP _{ag}	Aux*	<i>t</i>	<i>me-</i>	<i>t</i>	N-V	DP _{th}
	↑		↑		↑		
b. <i>Di</i> - passive:	DP _{th}	Aux*	<i>t</i>	<i>di-</i>		V	<i>t</i>
	↑		↑				↑
c. Bare passive:	DP _{th}	Aux*	<i>t</i>		DP _{ag}	V	<i>t</i>
	↑		↑				↑

The only difference between the standard languages and Desa in these basic clause types is in the realization of active voice verbs. Recall that active voice verbs in Desa may bear the long *meN*- or short *N*- prefixes, whereas active voice verbs in SI and SM are always introduced by the full form *meN*-.²⁰ For Desa, we proposed to treat *N*- as the realization of ν_{ACT} and *me-* as the optional realization of Voice in the context of

v_{ACT} . We follow Benjamin 2009 and Gil 2002 in proposing that the active form *meN-* reflects two prefixes, *me-* and *N-*, even in languages such as SI and SM where active verbs always appear with the full form, *meN-*. We present arguments for splitting *meN-* in this way in section 5.1 below.

Formally, we implement this result for SI and SM by proposing that both v_{ACT} and Voice realize their overt affix if and only if they are linearly adjacent to one another — see vocabulary items in (49ai, bi) — with null allomorphs as their elsewhere cases, in (49aiii, bii). These vocabulary items correctly predict that neither *me-* nor *N-* appear in isolation in SI and SM, unlike in Desa. We also adopt the vocabulary item for passive clauses in (49aai, c) from the Desa vocabulary items in (32) above.

(49) **Vocabulary items in SI/SM:**

- | | |
|--|---|
| <p>a. i. Voice $\leftrightarrow$ <i>me-</i> / ____ * v_{ACT}</p> <p>ii. Voice $\leftrightarrow$ <i>di-</i> / ____ * v_{PASS}</p> <p>iii. Voice $\leftrightarrow$ $\emptyset$ / elsewhere</p> | <p>b. i. v_{ACT} $\leftrightarrow$ <i>N-</i> / Voice * ____</p> <p>ii. v_{ACT} $\leftrightarrow$ $\emptyset$ / elsewhere</p> <p>c. v_{PASS} $\leftrightarrow$ $\emptyset$</p> |
|--|---|

Next we address patterns of A'-extraction in SI and SM. Recall that the subject is the only nominal that can be A'-extracted from each of the basic clause types, with the exception of object extraction with “*meN*-deletion”, which we discuss in the next section. The basic subject-oriented extraction restriction is explained by VoiceP being a phase. As introduced in section 3 above, phases can be described as being subject to the Phase Impenetrability Condition (PIC), which suggests that phrasal movement out of a phase must proceed through the phase head's specifiers, also described as the *phase edge*. As Voice always hosts exactly one nominal specifier, the subject, we predict that no other nominal can move out of VoiceP in the basic case. The subject that moves first to Spec,TP is the only nominal that can be A'-extracted from the clauses in (48). We accurately predict that non-nominal constituents, however, can move through the phase edge and therefore may undergo A'-extraction; see examples in (41) above.

²⁰ Some works note the possibility of “bare active” clauses, where the verb bears no voice prefix but the subject is high (i.e. preceding auxiliaries), in certain varieties of Indonesian and Malay. Chung (1978) call these “stem sentences”. As Voskuil (1996: 58 note 1) writes, “There is some controversy about the bare active; prescriptive and descriptive sources alike often claim that the construction is unacceptable or, at the very least, ‘sub-standard’”. For instance Tjung 2006 and Cole et al. 2008 describe the bare active as a distinct feature of certain non-standard, regional dialects. In contrast, Voskuil (1996, 2000) and Nomoto (2010, 2013) have argued that the bare active deserves to be described as a feature of the “Standard” languages, albeit being more common in colloquial and spoken registers.

Here we concentrate on addressing the consensus description for the “Standard” languages: active verbs must bear *meN-*, but it is possible to extract an object across a bare verb and a high (pre-auxiliary) subject, as we discuss in section 4.2 below. For varieties that regularly allow bare actives, we hypothesize that speakers may optionally access a grammar where both v_{ACT} and v_{ACT} -adjacent Voice are realized as $\emptyset$ -, resulting in their simultaneous null realization.

For completeness, we also note that Desa speakers judged bare active examples to be grammatical, but only volunteered active verb forms with *N-* or *meN-*. We note that this possible acceptability of bare actives in Desa does not affect our argumentation in the preceding sections. There is still a clear, three-way distinction in the range of verbal forms available, between actives (where *N-* and *meN-* are in free variation), bare passives (**N-*, **meN-*), and object extraction (*N-* ok but **meN-*), which motivates our proposal above.

In summary, we propose that the syntactic derivations for the basic clause types in SI and SM are identical to those in Desa, with only a minor difference in the morphological realization of active voice verbs. This proposal derives the basic subject-only A'-extraction restriction in the same way that we did for Desa, due to the phasehood of VoiceP.

4.2 Object extraction in SI/SM and Cyclic Linearization

We turn next to the derivation of object extraction clauses in SI/SM, which appears to violate the basic subject-only restriction on nominal A'-extraction. We begin by briefly reiterating the key features of object extraction clauses in SI/SM, illustrated with the topicalization example in (50). Object extraction clauses involve A'-extraction of a nominal that is not the local external argument. The external argument (*Badu* in (50)) appears in the high subject position, preceding any preverbal auxiliaries (*sudah*), just as in active clauses. The verb cannot bear a voice prefix, as in bare passive clauses but unlike active clauses.

(50) Object extraction in SI/SM has high agent and no voice prefix:

Buku ini Badu sudah { ***mem**-baca / baca } ____.
 book DEM Badu ASP ACT-read read

'This book, Badu has read.'

(SI; Voskuil 2000: 199, Musgrave 2001: 60)

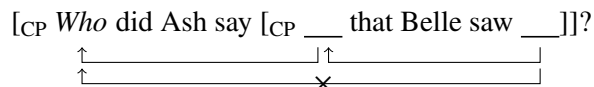
Against the backdrop of our discussion thus far, the most notable feature of SI/SM object extraction clauses is that *two nominals move simultaneously out of VoiceP*: Here, the theme *buku ini* A'-moves to Spec,CP and the agent *Badu* moves to the subject position, Spec,TP. This makes SI/SM object extraction clauses unlike all other clause types, and not derivable based on our proposal as presented so far. As discussed above, we propose that VoiceP always hosts exactly one nominal specifier and is a phase; assuming that only the specifiers of a phase can move out — due to the Phase Impenetrability Condition (PIC) of Chomsky 2000, 2001 — we then predict it to be impossible for multiple nominals to move simultaneously out of VoiceP.

We argue that we can derive the possibility of object extraction in SI/SM — including its obligatory absence of a voice prefix, as well as various other aspects of this construction, which we discuss in the remainder of section 4, below — from our overall proposal for Malayic syntax, by refining our characterization and understanding of phase impenetrability effects. We will continue to maintain that VoiceP constitutes a phase and always hosts exactly one nominal specifier.

Introducing Cyclic Linearization Recall that the postulation of phases reflects the intuition that certain syntactic units constitute natural boundaries with limited permeability. For instance, the phasehood of full clauses (CPs) has been proposed as underlying the observation that long-distance (cross-clausal) movement

proceeds in a successive-cyclic fashion (see e.g. Van Urk, 2020). An English example with the hypothesized intermediate movement step through the embedded CP edge indicated is repeated here in (51).²¹

(51) **Long-distance movement is successive-cyclic:** =(33)



So far in our discussion, we have made reference to Chomsky’s (2000: 108, 2001: 13) Phase Impenetrability Condition (PIC) as the relevant description of the opacity of phases. The PIC in (34) claims that only the “edge” of a phase (its head and specifiers) is accessible for syntactic operations from outside the phase, disallowing the “one-fell-swoop” movement as in (51).

Fox and Pesetsky (2005) develop a particular view of phasehood which has the effect of deriving many such restrictions on non-successive-cyclic movement, without having to rigidly assume Chomsky’s PIC as a condition on syntactic operations. Under their theory, dubbed *Cyclic Linearization*, after each full phase has been built, the structure undergoes *Spell-Out*, wherein the pronunciation and linearization of its contents are determined. The contents of the linearized phase can still be targeted by syntactic operations in a higher phase, including movement, as long as the relative ordering established by earlier *Spell-Out* is not contradicted.

Let us see how the theory of Cyclic Linearization serves to enforce the successive cyclicity of movement in (51). We again assume each CP to be a phase (see note 21 on the verbal phase). We first consider the ungrammatical derivation in (52), where *wh*-movement does not involve an intermediate step to the embedded clause edge. When the embedded CP is complete, *who* is still in the object position. At embedded CP *Spell-Out*, we therefore establish that *who* must follow the other constituents in the embedded CP; see (52a). We then build the matrix CP, including moving the embedded object *who* directly to the matrix Spec,CP. At matrix CP *Spell-Out*, we then establish that *who* must precede the other constituents in the matrix CP; see (52b). This yields an ordering contradiction: we established that *who* follows the other material in the embedded clause but also must precede it. The impossibility of coherently linearizing the entire structure leads to its ungrammaticality.

²¹ Assuming that maximal verbal projections (*vP* for Chomsky, but here *VoiceP*) are also phases, movement must proceed through the edge of these intermediate phases as well. For ease of presentation, we do not illustrate effects of the verbal phase in (51–53).

(52) **One-fell-swoop movement yields an ordering paradox:**

* [_{CP} *Who* did Ash say [_{CP} that Belle saw ____ ?


a. Linear order relations at embedded CP Spell-Out:

$C_{that} < DP_{Belle} < V_{saw} < DP_{who}$

b. Linear order relations at matrix CP Spell-Out:

$DP_{who} < C_{did} < DP_{Ash} < V_{say} < CP$

$\Rightarrow$ ordering paradox! ($DP_{who} < \dots < V_{say} < C_{that} < \dots < DP_{who}$)

We avoid this ordering paradox if *who* instead moves first to the edge of the embedded CP as in (53). Spell-Out of the embedded CP establishes that *who* precedes all other material in the embedded clause; see (53a). After the *wh*-word is moved further, Spell-Out of the matrix CP records that *who* precedes all other material there as well. Linearization of the entire structure succeeds in this case, resulting in the total ordering in (53c), explaining the need for successive cyclic movement.

(53) **Successive-cyclic movement avoids an ordering paradox:**

[_{CP} *Who* did Ash say [_{CP} ____ that Belle saw ____ ?


a. Linear order relations at embedded CP Spell-Out:

$DP_{who} < C_{that} < DP_{Belle} < V_{saw}$

b. Linear order relations at matrix CP Spell-Out:

$DP_{who} < C_{did} < DP_{Ash} < V_{say} < CP^{22}$

c. Total ordering possible:

$DP_{who} < C_{did} < DP_{Ash} < V_{say} < C_{that} < DP_{Belle} < V_{saw}$

Fox and Pesetsky (2005) argue that this theory of Cyclic Linearization explains apparent phase impenetrability effects on overt movement, and in fact is empirically superior to simply mandating through the PIC that only the phase edge is accessible for movement. What makes the edge of the phase special is not its structural position, but rather that it is linearly leftmost in the phase, and therefore its leftward movement will not contradict previous ordering relations. Further work that supports the Cyclic Linearization approach to phase impenetrability effects include Ko 2007, 2011, 2014, Sabbagh 2007, Medeiros 2013, Erlewine 2017, Davis 2020a,b, 2021, Lee 2021, and citations there.

²² See Fox and Pesetsky 2005, especially its appendix, for discussion of the linearization of trace positions of movement. Here it suffices to treat the precedence relation “< CP” as indicating precedence with respect to the unmoved parts of CP, and similarly for VoiceP in (56) below.

(54) **Object topicalization in SI/SM:** =(50)

(SI; Voskuil 2000: 199, Musgrave 2001: 60)

(55) **Object extraction VoiceP:**

[VoiceP DP_{A'} Voice [_{VP} DP_{ag} v_{ACT+V} t
Ø- Ø-*baça*

38

c. Linear order relations at VoiceP Spell-Out:

$$DP_{buku\ ini} < DP_{Badu} < v+V_{baca}$$

Next we build the higher phase of the clause. T moves the agent to Spec,TP, above the temporal auxiliary *sudah*, and licenses it. The C head above attracts the theme to Spec,CP. The complete derivation for (54) is thus reflected in (56) below. At CP Spell-Out, we establish the ordering statements in (56b).

(56) **Object extraction CP:**

a.
$$[_{CP} DP_{A'} C [_{TP} DP_{ag} \dots [_{VoiceP} t Voice [_{VP} t v_{ACT+V} t]]]$$

b. Linear order relations at CP Spell-Out:

$$DP_{buku\ ini} < DP_{Badu} < Aux_{sudah} < VoiceP$$

c. Total ordering possible:

$$DP_{buku\ ini} < DP_{Badu} < Aux_{sudah} < v+V_{baca}$$

The linearization statements established at VoiceP and CP Spell-Out are compatible with each other in this case, together producing the total ordering for the utterance in (56c). The movements of the theme and agent are order-preserving, requiring at each phase for *buku ini* to precede *Badu*. Furthermore, as these two constituents were the two leftmost parts of VoiceP as linearized in (55c), their further movement in (56) did not cross any overt material linearized in VoiceP which would have led to an ordering paradox. In particular, this simultaneous movement of the theme and agent was made possible by the null allomorph of Voice, independently motivated for the derivation of the bare passive. We discuss the importance of null Voice and its connection to the bare passive form over the next two sections.

4.3 On the importance of null Voice

The nature of the “*meN*-deletion” effect has received significant attention in previous theoretical work on SI and SM. Although many details differ, these prior proposals share the following intuition. The functional head associated with the voice prefix may bear different formal features, depending on whether a non-subject nominal A'-moves over it or not. (For some authors, this reflects the presence of a feature that triggers the movement, while others make reference to a feature that reflects that such movement has taken place.) The resulting feature bundle associated with movement of a non-subject is realized as a null allomorph. See proposals to this effect in Aldridge 2008b: 1456, Cole and Hermon 2005: 85–86, Cole et al. 2008: 1535, Sato 2012: 41–42, Georgi 2014: 151–156, Erlewine 2016b: 304–305, Jeoung 2017: 31, 2018a: 95–96, 2018b: 25,

and Keine and Zeijlstra 2025: 1174–1175. For these accounts, there is no deeper reason why the allomorph that appears in such cases is a null allomorph. The extraction-indicating allomorph could just as well be a different overt form, as is the case for Irish complementizer morphology (McCloskey, 1979, 2001) or Chamorro *wh*-agreement (Chung, 1982, 1998); see also Georgi 2017 for an overview of such morphology.

In contrast, our approach here directly predicts that Voice *must* be null to allow for object extraction to occur. To see why this is the case, let us consider what would happen if we build an object extraction clause following the proposal above as in (57a) below, but Voice then realizes an overt form such as *me-*. At VoiceP Spell-Out, we establish that *me-* precedes the agent DP in Spec,vP. In the construction of the CP, we move the agent DP to Spec,TP, as well as the nominal in Spec,VoiceP to Spec,CP, in an order-preserving fashion. However, CP Spell-Out determines that the moved agent DP must precede all non-moved parts of VoiceP, including *me-*. We are then left with two contradicting ordering statements — Voice both precedes and follows the agent DP — leading to the ungrammaticality of this structure. It is precisely this conflict which we avoid when the realization of Voice is null and thereby pruned.

(57) **Ungrammatical object extraction with overt Voice:**

- a. $[_{CP} DP_{A'} C [_{TP} DP_{ag} \dots [_{VoiceP} t \text{ Voice } [_{vP} t v_{ACT+V} t]]]$ =(56)
- $\uparrow$ $\uparrow$ $\uparrow$ $\uparrow$
- b. Linear order relations at VoiceP Spell-Out:
 $DP_{A'} < \text{Voice}_{me-} < DP_{ag} < v+V \dots$
- c. Linear order relations at CP Spell-Out:
 $DP_{A'} < DP_{ag} < \dots < \text{VoiceP}$
- $\Rightarrow$ ordering paradox! ($\underline{DP_{ag}} < \text{Voice}_{me-} < \underline{DP_{ag}}$)

We have furthermore proposed that the realization of Voice as *me-* is one-to-one with the realization of v_{ACT} as *N-* in SI and SM (see (49)), unlike in *Desa*. Therefore, in object extraction clauses in SI and SM, the verb must bear no voice prefix. In this way, our account explains why the “*meN*-deletion” interaction must take the form that it does: that is, taking advantage of the existence of a null allomorph for a functional head to facilitate movement.²⁴

²⁴ The consistent direction of this effect — a null allomorph allowing for a particular movement — is reminiscent of *complementizer-trace effects*; see Pesetsky 2017 for an overview. In a number of unrelated languages, including English (Perlmutter, 1968), Levantine Arabic (Kenstowicz, 1983, 1989), and Igbo (Amaechi and Georgi, 2019), an embedded clause’s subject can be A’-extracted across a null complementizer but not across an overt complementizer. See Erlewine 2017 for a proposal for such effects that builds on Cyclic Linearization, similar to our proposal here.

Support for our proposal comes from the behavior of a limited set of psych verbs such as *suka* ‘like’ that bear no prefix (see e.g. Sneddon, 1996: 267–268). The object of these verbs can also be extracted across a high subject. Notice that the external argument *saya* precedes the future auxiliary in example (58) below, indicating that both the focused object (or a corresponding relative operator; see note 23) and the external argument have moved out of VoiceP.

(58) **Object extraction from psych verb with high subject:**

Ini yang saya akan suka ____.

DEM C 1sg FUT like

‘This is the one that I will like.’

(SI; Stevens, 1970: 71)

The possibility of object extraction from these prefixless psych verbs as in (58) is compatible with our analysis. The Voice head associated with these verbs is generally null, and therefore allows for the simultaneous movement of the object (first moved to Spec,VoiceP) and the subject (generated in Spec,vP) out of VoiceP in an order-preserving manner.

These prefixless psych verbs teach us that the possibility of object extraction is not a special quirk of verbs that normally bear *meN-*, but instead reflect a more general possibility of Malayic clause structure, made possible in configurations where Voice is null. See also Soh 1998, 2013 and Fortin and Soh 2014 for an additional argument from transitive verbs that are optionally prefixed by *ber-*, which allow for object extraction only with their prefixless form. These interactions are precisely as predicted by our proposal and the Cyclic Linearization theory of phasehood.

Our proposal also offers a new understanding for the relationship between the bare passive and object extraction clauses. Although object extraction clauses cannot be simply reduced to instances of bare passives — due to differences in the position of their agents, as observed first by Soh (1998) and Cole and Hermon (2005) and reviewed in section 2.3 above — both require Voice to not be realized as an overt prefix. In bare passives (as well as object extraction clauses in Desa; see §3.3), if Voice were to be realized as an overt prefix such as *di-*, it would fail to affix to *v+V* via Local Dislocation, as the agent DP intervenes between the two heads. Under our analysis for SI and SM, we propose that overt allomorphs are inserted for Voice when linearly adjacent to *v*; see (49). The agent DP intervenes between Voice and *v+V* at VoiceP Spell-Out in both bare passives and object extraction clauses, correctly leading to the insertion of the null allomorph in these cases.

We provide further evidence for our account of the causal link between the null Voice allomorph and object extraction through discussion of two registers of Madurese, in section 5.3 below.

4.4 Further properties of object extraction

We now return to the syntax of object extraction in SI and SM. Following discussion in Soh 1998 and Cole and Hermon 1998, we have emphasized that two nominals move simultaneously out of VoiceP in these constructions: one to Spec,TP (the subject) and one to Spec,CP, as schematized in (59).

(59) **Moving two nominals out of VoiceP:**

$$[_{CP} \text{ DP } [_{TP} \text{ DP } \dots \text{ Aux}^* \dots [_{VoiceP} \dots \text{ (requiring no voice prefix)}]$$

Our proposal for such derivations makes very particular predictions about the conditions under which such a derivation would be grammatical, including that the Voice head must be unpronounced, as we discussed in the previous two sections. In this section, we discuss the choice of nominals that end up in Spec,CP and Spec,TP in such configurations, which provides further support for our proposal over other alternative approaches.

First, our proposal predicts that the configuration in (59) can only arise if the local external argument (generally an agent) moves to Spec,TP, and therefore the A'-extracted nominal in Spec,CP cannot itself be the local external argument. Concretely, this leads us to predict that the agent of a bare passive cannot be A'-extracted, as first noted by Chung (1976b: 85), thus explaining the ungrammaticality of structures such as (60).

(60) **Ungrammatical bare passive agent A'-extraction:**

* *Siapa_i yang pintu itu_j akan _____i buka _____j?*
 who C door that FUT open
 Intended: 'Who will the door be opened by?' (SI; based on Vamarasi, 1999: 55)

In order to move both *siapa* and *pintu itu* out of VoiceP as in (60), we would first move the theme *pintu itu* to Spec,VoiceP, establishing the relative order *pintu itu* < *siapa* at VoiceP Spell-Out. However, movement of *siapa* to Spec,CP and *pintu itu* to Spec,TP leads to the conflicting ordering *siapa* < *pintu itu* at CP Spell-Out. This explains the ungrammaticality of this structure.

The ungrammaticality of structures as in (60) serves as an argument against an alternative approach to "meN-deletion" effects. Suppose that we took the absence of overt Voice to indicate that there is no verbal phase boundary (either with the Voice head not projected or lacking its status as a phase head), predicting all nominal arguments to be accessible for movement. Such a theory would not predict the attested restriction on the order of moved nominal arguments, which is correctly predicted by our Cyclic Linearization account.

Next we discuss the choice of nominal that undergoes A'-movement to Spec,CP in (59). Although in most cases above we have presented examples that A'-move the local direct object, in fact a range of different

nominals can be A'-extracted, leading to the configuration in (59). For example, Sato (2012) shows that A'-extraction can target either argument of certain ditransitives as in (61) and Jeoung (2018b) shows that A'-extraction can target the possessors of objects as in (62). (See Jeoung 2018b for arguments that such examples involve A'-movement of the possessor, rather than another relation such as pronominal binding.) A'-extraction can also target the arguments of embedded clauses, as in (63–64) below. In all of these examples, verbs that are along the path of A'-movement cannot bear their voice prefix.

(61) **A'-extraction of ditransitive objects:**

a. *Apa yang kamu { *mem-beli-kan / beli-kan } ibu-mu ____.*
 what C you ACT-buy-APPL buy-APPL mother-your
 'What did you buy your mother?'

b. *Siapa yang kamu { *mem-beli-kan / beli-kan } ____ bunga.*
 what C you ACT-buy-APPL buy-APPL flower
 'Who did you buy flowers?'

(SI; Sato, 2012: 43)

(62) **A'-extraction of object possessor:**

*Siapa yang adik { *mem-baca / baca } [buku-nya ____]?*
 who C younger.sibling ACT-read read book-D
 'Whose book is that little brother reading?'

(Colloquial Ind.; Jeoung, 2018b: 10)

(63) **A'-extraction of embedded subject:**

*Apa-kah yang mereka { *ber-harap / harap } [CP ____ akan di-terima]?*
 what-Q C 3pl MID-hope hope FUT PASS-accept
 'What do they hope will be accepted?'

(SM; Soh, 1998: 305)

(64) **A'-extraction of embedded object:**

*Siapa yang Bill { *meng-ira / kira } [CP Fred { *men-cintai / cintai } ____]?*
 who that Bill ACT-think think Fred ACT-love love
 'Who does Bill think Fred loves?'

(SI; based on Saddy, 1991: 187)

As previously noted by Aldridge (2008b), Sato (2012), and Jeoung (2018b), such examples argue against any approach to non-subject extraction in SI/SM that specifically ties the absence of a voice prefix to the extraction of the direct object or theme, as in Cole and Hermon's (2005) case agreement account. The A'-extracted nominal may not even be an argument of the verb, as it may be a possessor (62), an embedded subject (63), or embedded object (64). By the same token, what we have called "object extraction" clauses above, then, may more accurately be described as the general structure for the A'-extraction of any nominal that is not the local external argument from within VoiceP.

Finally, we note that non-nominal phrases can move long-distance, crossing Voice prefixed verbs in each clause. See example (65) below, the interpretation of which requires ‘to who’ to originate as a PP argument of the embedded verb ‘give.’ Such examples are straightforwardly derived on our account, as VoiceP can host the subject nominal and a non-nominal simultaneously as specifiers, allowing for both to move out of the phase while realizing the active prefix; see schema in (42).

(65) **A'-extraction of embedded PP:**

Kepada siapa kamu **meng**-harap-kan [_{CP} Ali akan **mem**-beri mobil-nya ____]?
to who 2sg meng-hope-APPL Ali FUT ACT-give car-3sg
‘To whom do you hope Ali will give his car?’ (SI; Fortin and Soh, 2014: 50)

In summary, our proposal correctly predicts that object extraction clauses as schematized in (59) above allow for any non-subject to land in Spec,CP, with the local external argument in Spec,TP. Under our Cyclic Linearization account, the only way that multiple nominals can move out of the VoiceP phase is if they are a non-subject and the local external argument, in that linear order, with no overt realization of Voice.

4.5 Object extraction and ellipsis

Further evidence for our proposal comes from the interaction of object extraction with ellipsis. (We thank Norvin Richards (p.c.) for encouraging us to consider such interactions.) Fortin (2007) shows that Standard Indonesian allows for VP ellipsis under certain auxiliaries, but that voice mismatches are disallowed (see especially pages 269–272). Now consider example (66):

(66) **Grammatical object extraction from elided VP with active antecedent:**

Ali bisa mem-beli sepeda. Apa yang Siti bisa Δ ?
Ali can ACT-buy bike what C Siti can
‘Ali can buy a bike. What can Siti (buy)?’ (SI; Fortin, 2007: 265)

The second clause in example (66) is an object extraction clause, with A'-movement of the theme *apa* ‘what’ and the agent *Siti* in the high, pre-auxiliary position. It also involves VP ellipsis (referring pre-theoretically to ellipsis of an extended verb phrase constituent), with the ellipsis site indicated by Δ interpreted under identity with the antecedent VP, *membeli sepeda* ‘buy bike’, modulo object extraction. (VP ellipsis of this form is also possible in English, as reflected in the translation.) Notice that a variant of this sentence without VP ellipsis would be ungrammatical if realized with the string-identical *membeli*, although the prefixless *beli* is possible.

The grammaticality of structures of the form in (66) is predicted by our proposal. Much prior work has shown that ellipsis requires a structurally identical linguistic antecedent, modulo material that is focused and/or

moved out of the ellipsis site. In particular, Merchant (2013) argues specifically on the basis of interactions between voice alternations and VP ellipsis in English and other languages that “the identity relation between an elided phrase XP_E and its antecedent XP'_A is one of syntactic featural identity” (p. 88), not requiring identity of surface forms. On our proposal, the active antecedent clause and the object extraction clause in (66) involve featurally identical functional heads, Voice and v_{ACT} , despite differences in the derivations, as illustrated in (67) below.

- (67) a. [TP DP_{ag} ... Aux ... [VoiceP *t* Voice [vP *t* v_{ACT+V} DP_{th}]
↑ ↑
Ali bisa me- N-beli sepeda
- b. [CP DP_{A'} C [TP DP_{ag} ... Aux[E] ... [VoiceP *t* Voice [vP *t* v_{ACT+V} *t*]
↑ ↑ ↑
apa yang Siti bisa Δ

As per the proposals in Fortin 2007, Merchant 2013, and citations there, we suppose that the auxiliary *bisa* ‘can’ may bear the ellipsis feature [E] in (67b), triggering PF deletion (or the lack of vocabulary insertion) for its complement. We therefore predict the grammaticality of (66), despite the fact that the surface form for the elided verb — if pronounced — would differ in form from the antecedent verb. The active verb with *meN-* and prefixless verb in object extractions differ on our account only in their morphological realizations, not in the syntactic heads involved. In turn, both featurally differ from passives (with v_{PASS}), explaining the unavailability of active–passive voice mismatches between antecedent and elided VPs (Fortin, 2007: 269–272).

5 Voice prefix structure and variation

In this section, we take a closer look at the organization of voice prefixes and their cross-linguistic variation. In section 5.1, we present arguments for the idea that the active formative *meN-* in SI and SM reflects two prefixes, *me-* and *N-*, as we had assumed in our analysis above. We also discuss the extension of our proposal to a range of regional and colloquial Malayic varieties where active verbs exhibit only a homorganic nasal, in section 5.2. Finally, in section 5.3, we discuss voice and extraction patterns in two varieties of Madurese, which provides further support for our account.

5.1 Splitting active *meN-* in Standard Indonesian and Malay

An important component of our proposal for Malayic clausal syntax is the idea that the extended projection of the verb involves two functional heads, Voice and v , following Nomoto 2015, 2021, and in particular that

these two heads correspond to two separate prefixes that appear on active verbs in SI/SM, *me-* and *N-*. We motivated such a decomposition first for Desa, where the active prefixes *N-* and *meN-* are generally in free variation but the behavior of object extraction clauses motivates treating the *me-* component separately. For Malayic languages such as SI and SM where active verbs always realize the full prefix *meN-*, the proposed split is less obvious and warrants further discussion and motivation.

The idea that active verb forms with *meN-* in SI and SM reflect two prefixes, *me-* and *N-*, is suggested by Benjamin (2009) and Gil (2002). Together with these authors, we note that there are multiple processes in these languages that would lead a learner to treat the *meN-* of active verbs as the combination of two prefixes, *me-* and *N-*. Here we highlight two such considerations: parallels between *meN-* verbs and *peN-* nominalizations and the behavior of *meN-* verbs under reduplication.

The first process that motivates the decomposition of *meN-* is deverbal nominalization with *peN-* versus *pe-*. The phonological realization of the nasal *N* in *peN-* nominalizations is identical to that of the nasal *N* in *meN-* active verbs (see e.g. Sneddon, 1996: 9–14), and the phonological alternation is not a general process in these languages (Blust, 2004). Although the range of stems that take *peN-* versus *pe-* and their resulting denotations are subject to individual lexical idiosyncracies, there nonetheless are a number of established generalizations that support treating *peN-* nominals as tightly linked to *meN-* active verbs. We argue with Benjamin (2009: 303–304) that such distributional and semantic correspondences between *peN-* and *meN-* support their decomposition into *pe-* + *N-* and *me-* + *N-*, with *N-* being a shared agent/causer-introducing functional head: that is, what we call v_{ACT} .

Semantically, prior works observe that *peN-* nominals denote the subject (agent/causer) or instrument of the corresponding *meN-* active verb,²⁵ whereas *pe-* nominals correspond to the subject of *ber-* middle verbs or, in some cases, the subject of a *di-* passive. See Sneddon 1996: 27–30, Dardjowidjojo 1978: 277–278, and various citations in Denistia 2018, as well as corpus evidence in Denistia and Baayen 2019 that supports this correlation. *Ber*-prefixed middle verbs are intransitive or transitive predicates with various interpretational restrictions, depending on the stem; see for example Kemmer 1993, Wee 1995, Wong 1998, Benjamin 2009, Beavers and Udayana 2023, Karaj and Sansò 2023, and citations there.

A limited set of verbal stems appear with both *peN-* and *pe-*, for which two generalizations have been noted in the literature. First, for one class of predicates, the resulting *peN-* nominals are agent/causer-denoting whereas corresponding *pe-* nominals are theme-denoting, as exemplified in (68):

²⁵ Stems vary idiosyncratically in what variety of *peN-* nominal they derive. *PeN-* nominals with certain stems are ambiguous between agent and instrument readings, such as with *pen-cetak* ‘printer’ (Sneddon, 1996: 28) or *peng-gerak* ‘mover’ (Denistia and Baayen, 2019: 395), just as their English translations are as well.

(68) **Some agentive *peN*- vs patientive *pe*- nominalizations:**

(from Benjamin 2009: 304 crediting Hassan 1974: 59–63; Sneddon 1996: 29–30; Denistia 2018)

- | | | | |
|----|----------------------|------------------------------|--|
| a. | <i>sakit</i> ‘sick’ | <i>peny-akit</i> ‘disease’ | <i>pe-sakit</i> ‘patient’ |
| b. | <i>sapa</i> ‘greet’ | <i>peny-apa</i> ‘greeter’ | <i>pe-sapa</i> ‘addressee’ |
| c. | <i>suruh</i> ‘order’ | <i>peny-urut</i> ‘commander’ | <i>pe-suruh</i> ‘one who takes orders’ |
| e. | <i>tatar</i> ‘train’ | <i>pen-atar</i> ‘trainer’ | <i>pe-tatar</i> ‘trainee’ |
| f. | <i>tunjuk</i> ‘show’ | <i>pen-unjuk</i> ‘indicator’ | <i>pe-tunjuk</i> ‘instructions’ |
| g. | <i>ubah</i> ‘change’ | <i>peng-ubah</i> ‘changer’ | <i>pe-ubah</i> ‘variable’ |

Second, for another class of predicates, both *peN*- and *pe*- nominals are at first glance both agent-denoting, but subtly differ. The *peN*- forms are straightforward agent nominalizations of the activity described by the verb stem, whereas the corresponding *pe*- nominals appear to be what Zobel (2017) calls *role nouns*, which are conventionally associated with the activity but conceptually richer (see also Maienborn, 2020). (We thank Ryan Walter Smith (p.c.) for discussion here.)

(69) **Some *peN*- agent nominalizations vs *pe*- role nouns:**

(from Sneddon 1996: 30; Denistia 2018)

- | | | | |
|----|------------------------------|--|-------------------------------|
| a. | <i>jabat</i> ‘(hold) office’ | <i>pen-jabat</i> ‘acting’ (i.e. temporary) | <i>pe-jabat</i> ‘official’ |
| b. | <i>selam</i> ‘dive’ | <i>peny-elam</i> ‘one who dives’ | <i>pe-selam</i> ‘diver’ |
| c. | <i>serta</i> ‘accompany’ | <i>peny-erta</i> ‘accompanying’ | <i>pe-serta</i> ‘participant’ |
| d. | <i>tembak</i> ‘shoot’ | <i>pen-embak</i> ‘one who shoots’ | <i>pe-tembak</i> ‘shooter’ |
| e. | <i>tinju</i> ‘punch’ | <i>pen-inju</i> ‘one who punches’ | <i>pe-tinju</i> ‘boxer’ |

These observed generalizations regarding the use of *peN*- versus *pe*- further support the identification of a shared *N*- between *peN*- nominals and *meN*- active verb forms. As we see, *peN*- nominals regularly denote agent nominalizations, directly corresponding to the subject of the corresponding *meN*- verb form. In contrast, *pe*- nominals may describe a wider range of types of nouns associated with the verb, in a more idiosyncratic fashion depending on the stem.

On our analysis, *meN*- active verbs necessarily introduce agent semantics through its v_{ACT} , whereas passives with v_{PASS} and *ber*- middles only optionally project an agent argument.²⁶ *PeN*- nominals include the v_{ACT} head of active verbs, whereas *pe*- nominals contain a different verbal structure, without v_{ACT} . These

²⁶ We do not develop a full account of the syntax/semantics of *ber*- verb constructions here. For our purposes, it suffices to note that, according to Beavers and Udayana (2023), a *ber*- middle can project an agent argument but nonetheless fails to (Case) license its internal argument. We conclude that the v head in *ber*- verbs cannot be v_{ACT} , which *does* license an internal argument.

correspondences would support a learner identifying a shared *N-* between *meN-* and *peN-*, with its shared semantics and identical phonological realization.

The second process that motivates splitting *meN-* is its interaction with verbal reduplication. SI and SM have a process which is commonly described as verbal stem reduplication, with various semantic functions (Sneddon, 1996: 20–21). Notably, when an active verb undergoes reduplication and so-called *nasal substitution* takes place, replacing the stem-initial consonant, both instances of the stem reflect nasal substitution but the syllabic, *me-* part of *meN-* appears only once (Onn, 1976; Lapoliwa, 1981; Sneddon, 1996: 14). See example (70), where we underline the effects of nasal substitution.

(70) **Verbal reduplication reflects nasal substitution but not *me* of active *meN-*:**

tulis ‘write’ > active *menulis* > *menulis-nulis* ‘write repeatedly’

(cf **menulis-tulis*, **menulis-menulis*)

This interaction has received substantial attention in the phonological literature, including as a motivating example for the idea of Base-Reduplicant faithfulness constraints in Optimality Theory, with McCarthy and Prince (1995: 86) describing it as “an otherwise unexplained phenomenon.” However, Benjamin (2009: 298) suggests that such interactions may be more straightforward if we posit nasal-substituting *N-* as a prefix that applies first, feeding reduplication, with *me-* being a separate, second prefix. In relation to our syntactic proposal, we may hypothesize that verbal reduplication targets the material in the head *v* (including the verb root which head-moves to *v*), to which the realization of Voice combines via Local Dislocation, although this approach is not without its own complexities.²⁷ But even setting aside the question of the best analysis of these facts (see note 27), surface patterns such as in (70) again provide a hint for the learner that supports the decomposition of *meN-* into two parts: *me-* and *N-*.

Finally, we note that the idea that Malayic *meN-* may reflect two prefixes echoes a long-standing hypothesis in Austronesian historical linguistics regarding the proper reconstruction of **maŋ-* in Proto-Malayo-Polynesian

²⁷ A complication to this account, which Benjamin does not discuss, is the fact that nasal substitution only applies to stems with initial voiceless consonants, and non-substituting *N* is not targeted by reduplication. For example: *baca* ‘read’ > active *membaca* > *membaca-baca* ‘read repeatedly’ (cf **membaca-mbaca*). The approach to reduplication that we suggest here would require adjusting our vocabulary items so that *N* reflects *v_{ACT}* only where the verb stem has an initial voiceless consonant, with Voice realizing *meN-* instead of *me-* otherwise. Such an amendment to our vocabulary items is possible, as a straightforward instance of inwards-sensitive phonologically conditioned allomorphy (see e.g. Bobaljik, 2000).

Support for the plausibility of such an approach comes from the fact that, in certain colloquial varieties such as Johor Malay as reported in Onn 1976, with stem-initial consonants which are *not* targeted for nasal substitution, the nasal part of *meN-* is optionally included in verbal reduplication. For example, in Johor Malay: *gali* ‘dig’ > active *menggali* > *menggali-(ng)gali* ‘dig continuously’ (p. 178). These variants reflect optionality in treating *N* as the realization of *v_{ACT}* or as part of Voice in non-substitution contexts.

See also Blust 2004, whose discussion of the proper analysis of nasal substitution in SI/SM *meN-* and *peN-* concludes that “it is perhaps safest to say that the morpheme boundary does not properly precede or follow the nasal”, for reasons independent of the reduplication facts. As a matter of notation, he suggests “placing a morpheme boundary on either side of the prefixal nasal”: that is, as *me-N-* (p. 83).

(PMP). As Blust (2004) notes, Dempwolff (1934: 30) and Tharp (1974: 1) suggest that “**maŋ-* was historically composite, consisting of morphemes **ma-* and **ŋ-*” (p. 90); see also discussion on pp. 111–112. We reiterate that our primary claim is about the synchronic representation of active voice forms in SI/SM, rather than their historical source. Nonetheless, these phonological considerations would also be relevant for the language learner, regardless of the form’s etymology.

5.2 *N-* active forms

In this section, we briefly discuss the availability of object extraction in other varieties of Malay/Indonesian and related languages of the region. In many such colloquial and regional varieties, active verb forms bear a shorter active prefix such as *N-* rather than the full *meN-* form of the standard languages (see e.g. Gil, 2002; Cole et al., 2008). When we consider the possibility of object A’-extraction in these varieties, two patterns of behavior emerge.

The first pattern, exemplified here by Jakarta Indonesian, parallels the “*meN*-deletion” interaction between voice morphology and extraction in SI/SM above. In Jakarta Indonesian, active verbs optionally bear the *N-* active prefix, as in (71a). A’-extraction of the subject is possible from an active clause with or without the prefix, as in (71b), as well as from other, non-active clause types (not shown here). A’-extraction of an object is possible only if the verb bears no prefix, as in (71c) below.

(71) **Extraction from Jakarta Indonesian actives:**

(Tjung, 2006: 22–24)

a. *N-* and bare actives:

Anak itu { **nge**-baca / baca } buku.
 child that N-read read book
 ‘The child is reading a book.’

b. Subject extraction:

Siapa yang ____ { **nge**-baca / baca } buku?
 who C N-read read book
 ‘Who is reading a book?’

c. Object extraction:

Apa yang anak itu { ***nge**-baca / baca } ____?
 who C child that N-read read
 ‘What is the child reading?’

Like SI and SM, Jakarta Indonesian also has bare passives with immediately preverbal agents. As Tjung (2006) shows, A’-extraction of a theme can cross a high subject that precedes auxiliaries as in (72), indicating

that the possibility of theme extraction across a bare verb cannot be reduced to the existence of bare passives (see also Cole et al., 2006).

(72) **Subject can be high in Jakarta Indonesian object extraction:**

Buku [_{RC} yang gua nggak lagi baca ____] mahal.
book C 1sg NEG PROG read expensive

‘The book that I am not reading is expensive.’

(*ibid.*: 25)

The same pattern of behavior — object extraction that is possible when the active prefix *N-* is absent, with a high subject that precedes auxiliaries — is also attested in Sarang Lan Malay (Cole et al., 2008: 1524–1527) and three dialects of Jambi Malay (Yanti, 2010: Tanjung Raden, pp. 46–47; Jambi City, pp. 52–53; Mudung Darat, pp. 58–60). It is also attested in some non-Malayic languages of the region with similar, so-called Indonesian-type syntax, including Semarang Javanese (Cole et al., 1999b), Kendal Javanese (Sato, 2012), and polite Madurese (Jeoung, 2017; see section 5.3 below).

For these languages, we propose that the active prefix *N-* is a realization of the head Voice. Adopting our overall proposal for the Malayic VoiceP above, our Cyclic Linearization analysis explains the possibility of object A'-extraction and simultaneous subject raising being dependent on the non-realization of Voice as *N-*.

The second possibility is that lone *N-* is the realization of *v*, similar to our proposal for Desa and SI/SM, but with Voice always being null. In this case, we predict the availability of object extraction across a *N-*-prefixed active verb. This pattern of behavior is attested in Kuching Malay as described in Cole et al. 2008. Object extraction is possible whether across a *N-*-prefixed active verb or a bare active, as we see in (73):²⁸

(73) **Object extraction in Kuching Malay:**

Apa (nok) kitak { *n-anam* / *tanam* } ____?
what C 2 N-plant plant

‘What did you plant?’

(Cole et al., 2008: 1548–1549)

Additional evidence for the possibility of associating lone *N-* with the lower head *v* rather than the higher head Voice comes from the Kendayan group of Malayic languages in western Borneo. In these languages, *N-*

²⁸ Cole, Hermon, and Yanti (2008) propose that nasal-initial forms in the language are synchronically analyzed as variant stem forms; that is, *nanam* and *tanam* are simply allomorphs for the stem ‘plant’ in free variation. However, the discussion there suggests that ‘plant’ is one of a class of verbs which allow for both *N*-initial and *N*-less variants. Positing an optional *N-* prefix for these forms, as we suggest here, is cognitively plausible depending on the distribution and frequency of such items (see e.g. Yang, 2016), and thus cannot be categorically ruled out.

Musgrave (2001) suggests that this pattern might also be identified in other colloquial Malay/Indonesian varieties as well: “Umar Muslim (p.c.) points out that in this case [in varieties where active verbs bear *N-* alone], the verb behaves like the bare verbs discussed in the preceding sections [where object extraction is described], and not like a prefixed verb” (p. 62 note 24). Unfortunately, however, he does not include supporting examples or name the specific varieties which exhibit this behavior.

appears not only on active verbs but also on passive verbs under certain conditions, as in (74a).²⁹ In addition, passive agents may intervene between *di*- and *N*- (Dunselman, 1949: 70; Kalom and Hudson, 1970: 287–288; Adelaar, 1992: 161–162 *et seq*), as in (74b). See also Sommerlot 2024 for evidence that these properties, exemplified here in (74) in Salako, indeed hold across the Kendayan subgroup.

(74) “*di* (agent) *N*-V” passives in Salako (Kendayan Malayic):

- a. Uma-e akà’ **di**-nga-rumput.
field-poss.3 done PASS-N-weed
‘Her field was already weeded.’
- b. Ang-koà-lah tuàkng kaleng **di**=kau **m**-atàh-**m**-atàh ang-koà. (*N*-patàh > matàh³⁰)
REL-DEM-EMPH bone catfish PASS=2sg N-break-RED REL-DEM
‘That’s the catfish-bone you’ve broken into many pieces.’

(Adelaar, 2002: 15, 18, 2005c: 218, 2005b: 55, 57)

The co-occurrence of *di*- and *N*- as in (74) as well as the position of the agent between *di* and *N*- in (74b) are easily accounted for by the organization of the verbal extended projection with Voice and *v* put forward by Nomoto (2015, 2021) and adopted here. It appears that in Kendayan, unlike in most other Malayic varieties, *N*- realizes a variety of the lower head *v* that has been reanalyzed with separate semantics (see note 29) and thus may cooccur with passive voice. In addition, with the *di* form of Voice reclassified as a proclitic instead of as a prefix, the word order in (74b) is explained by the agent being introduced as a specifier of the lower head *v*.

With *N*- realizing the lower head *v*, following our analysis of object extraction above, we predict that Kendayan languages will allow for object extraction across *N*- prefixed verbs. Example (75) shows that our prediction is borne out in Belangin, another Kendayan dialect (Adelaar, 2006; Sommerlot, 2024). Notice that the agent subject *aku* precedes the auxiliary *udah* in (75), indicating that this is an object extraction clause rather than a passive with subject extraction.

(75) **Object extraction in Belangin (Kendayan Malayic):**

(second author field notes)

- Motor [RC nang aku udah **m**-injam ____] ada kiya. (*N*-pinjam > minjam)
motorcycle C 1sg ASP N-lend exist here
‘The motorcycle that I lent out is here.’

²⁹ *N*- in passive voice indicates perfective aspect in some Kendayan varieties including Salako (Adelaar, 2005b: 57–61; Kaufman 2007: 629), but appears to be used more freely in other Kendayan varieties (Sommerlot, 2024). The cooccurrence of *di*- and *N*- in passives is also attested in Riau Indonesian (Gil, 2002), but without the possibility of agents intervening.

³⁰ Notice that the verb is reduplicated in (74b). Just as we suggested for SI/SM verbal reduplication in section 5.1 above, reduplication targets precisely the material we propose as realized in the *v* head, without the realization of Voice (here: *di*-).

In summary, our proposal for Malayic clausal syntax and its verbal functional projections based on Nomoto 2015, 2021 can account for a wide range of attested grammatical behaviors in lesser studied Malay and Malayic varieties. In particular, those with active verbs that bear *N-* alone may be analyzed as realizing *N-* on Voice or *v*, then correlating with the (im)possibility of object extraction across *N*-prefixed verbs. Prior accounts which posit a single verbal functional head face greater challenges in modeling the space of attested variation.

5.3 Evidence from Madurese register variation

Further evidence for our approach to the importance of null Voice for object extraction, as well as the relation between object extraction and bare passives, comes from Jeoung’s (2017) discussion of the familiar and polite registers of Madurese, another language of the region with so-called Indonesian-type syntax. Madurese is classified as non-Malayic but in the larger Malayo-Sumbawan group (Adelaar, 2005a). Earlier work on Madurese had described the language as having exactly two voices, which are both morphologically marked: active verbs bear a homorganic nasal *N-* (or *a-*; see note 31 below) and passive verbs bear the prefix *e-*. These two voices neatly parallel the behavior of the active voice versus *di-* passive in the Malayic languages that we introduced in section 2.2 above. Jeoung however demonstrates that polite register Madurese additionally has a bare passive, unlike in the familiar register, hypothesizing that previous descriptions were based primarily on work on the familiar register only (p. 17). The following examples illustrate these voice possibilities with the stem ‘call’, which is *tembhal* in the polite register and *kato* in the familiar register.

(76) **Three voices in polite Madurese:**

(Jeoung, 2017: 17, 20)

a. Active:

Ramah ampon n-embhal-ih potra-epon.
 father ASP ACT-call-APPL SON-DEF
 ‘Father called his son.’

b. *e-* passive:

Potra-epon ampon e-tembhal-ih sareng ramah.
 SON-DEF ASP PASS-call-APPL by father
 ‘The son was called by father.’

c. Bare passive:

Potra-epon ampon ramah tembhal-ih.
 SON-DEF ASP father call-APPL
 ‘Father called his son.’

(77) **Only two voices in familiar Madurese:**

(*ibid.*: 16, 25)

a. Active:

Ali ng-ato-eh ana'-eng.
Ali ACT-call-APPL child-DEF
'Ali called his child.'

b. e- passive:

Ana'-eng e-kato-eh bi' Ali.
child-DEF PASS-call-APPL by Ali
'The child was called by Ali.'

c. Bare passive:

* Ana'-eng la engko' kato-eh.
child-DEF ASP 1sg call-APPL
Intended: 'I called his child.'

Just as in the bare passive of Desa or SI/SM above, the agent in the polite register's bare passive must be immediately preverbal, following any auxiliaries:

(78) **Bare passive agent must be low in polite Madurese:**

Buku panekah { *ramah } ampon { ramah } bacah.
book that ASP father read
'Father already read that book.'

(*ibid.*: 22)

Jeoung also shows that the subjects of all of these clause types can be A'-extracted; see pp. 28–29.

In addition to the availability of the bare passive, Jeoung documents one other difference between the polite and familiar registers of Madurese: object extraction is possible in the polite register across a verb without its voice prefix (79), but not in the familiar register (80).³¹

(79) **Object extraction in polite Madurese:**

Buku panekah se ramah ampon { ***m**-acah / bacah } ____.
book that C father ASP ACT-read read
'It's that book that father read.'

(*ibid.*: 29)

³¹ Certain stems take *a-* instead of *N-* in its active form (see Davies, 2010: 99–102, 250–255). Jeoung (2016: 58) reports that verbs that take an *a-* active prefix also allow for object extraction by dropping the *a-* prefix; see exx. 23b vs 25b there.

(80) **Attempted object extraction in familiar Madurese:**

* *Buku jiyah* se David la {**m**-acah / bacah} ____.

book that C David ASP ACT-read read

Intended: ‘It’s that book that David read.’

(*ibid.*: 29)

As Jeoung points out, in the grammatical object extraction example in (79), the agent *ramah* ‘father’ precedes the auxiliary *ampon*. This is unlike agents in the polite register’s bare passive, which must be immediately preverbal; see (78) above. Therefore the structure in (79) cannot be described as an instance of bare passive subject extraction. Parallel to object extraction clauses in SI/SM, then, the agent in (79) is in the high, canonical subject position (Spec,TP) as in an active clause, but the verb bears no voice prefix as in a bare passive, inviting a description parallel to that of “*meN*-deletion” for SI and SM. Jeoung therefore concludes (pp. 30–31) that there are two separate points of variation between the polite and familiar registers in the inventory of functional heads: in the terms of her analysis, the polite register has an extra bare passive Voice head, as well as a variant active Voice head that is unpronounced and allows for object extraction. In other words, there is no direct causal relationship between these two contrasts.

We argue instead that the differences in the grammars of polite versus familiar Madurese, introduced above, can all be attributed to just one difference in the morphological realization of Voice: polite Madurese has a null allomorph for Voice, but familiar Madurese does not. First, given the overall parallels in their behaviors (described in detail in Jeoung 2017), we adopt our proposal for the syntax of the four clause types in Malayic — the three basic clause types as well as object extraction — for Madurese. We repeat our schemata for the derivation of these clause types in (81) below, with minor adjustments for the shape of voice prefixes. We propose that all four of these clause types can in principle be derived in the syntax of both polite and familiar Madurese, but that the bare passive (81c) and object extraction clauses (81d) fail to result in licit surface forms in the familiar register.

(81) **Deriving four clause types in Madurese:**

(based on (48, 56))

	[CP	[TP	...	[VoiceP	Voice	[_v P	<i>v</i> +V
a. Active:		DP _{ag}	Aux*		<i>t</i> N-	<i>t</i>	V DP _{th}
		↑			↑		
b. <i>E</i> - passive:		DP _{th}	Aux*		<i>t</i> <i>e</i> -		V <i>t</i>
		↑			↑		
c. Bare passive:		DP _{th}	Aux*		<i>t</i>	DP _{ag}	V <i>t</i>
		↑			↑		
d. Object extraction:	DP _{A'}	DP _{ag}	Aux*		<i>t</i>	<i>t</i>	V <i>t</i>
	↑	↑			↑		

As reflected in (81) above, for Madurese, we treat both *v*_{ACT} and *v*_{PASS} as null. A possible account of

these facts is as follows. In polite Madurese, Voice realizes the overt active and passive prefixes when linearly adjacent to the heads v_{ACT} and v_{PASS} , respectively, and is null otherwise (see (82)), just as we proposed for SI and SM above (see (49)). In contrast, in familiar Madurese, Voice realizes the overt active and passive prefixes when *structurally* adjacent to $v\text{P}$ headed by a particular flavor of v , as in (83) below.³²

(82) **Vocabulary items in polite Madurese:**

- a. Voice $\leftrightarrow N-$ / ____ * v_{ACT}
- b. Voice $\leftrightarrow e-$ / ____ * v_{PASS}
- c. Voice $\leftrightarrow \emptyset$ / elsewhere

(83) **Vocabulary items in familiar Madurese:**

- a. Voice $\leftrightarrow N-$ / ____ $v_{\text{ACT}}\text{P}$
- b. Voice $\leftrightarrow e-$ / ____ $v_{\text{PASS}}\text{P}$

Now consider how Voice will be realized when the VoiceP phase undergoes Spell-Out in each of the structures in (81) above. In the active and $e-$ passive clauses, Voice will be linearly adjacent to v — because the agent has already moved out of the way in (81a) or none was generated in (81b) — as well as a sister to its $v\text{P}$ projection, thereby allowing for Voice to realize its active or passive prefixes as per (82a,b) and (83a,b). These prefixes then affix to $v+V$ via Local Dislocation. Davies (2010: 256) and Jeoung (2017: 19, 33) note that these active and passive prefixes are obligatory in these clause types.

In the bare passive and object extraction clause types, the agent intervenes between Voice and v at the point of VoiceP Spell-Out. We propose that polite Madurese has an allomorph that can be realized in this circumstance (see (82c)), which furthermore is null and will therefore be pruned (Embick, 2003, 2010). Unlike an overt Voice prefix, the null Voice allomorph does not require Voice to be linearly adjacent to $v+V$ in order to affix to it via Local Dislocation, allowing for the bare passive with its in-situ agent in (81c), nor will it introduce conflicting ordering statements when the agent moves across it in an object extraction clause as in (81d). In contrast, the vocabulary items for familiar Madurese do not rely on linear adjacency with v , and therefore will continue to realize their overt prefix forms. In the bare passive (81c), the overt prefix will fail to affix to the verb in $v+V$. In the object extraction clause (81d), the overt affix will introduce contradictory ordering statements with respect to the linear position of the agent. We therefore predict that the vocabulary items in (83) lead to the ungrammaticality of the bare passive and object extraction constructions in the familiar register, unlike in the polite register.

³² Alternatively, one could posit two distinct flavors of Voice in familiar Madurese (Voice_{ACT} $\leftrightarrow N-$; Voice_{PASS} $\leftrightarrow e-$), which are in a selectional relationship with their corresponding v heads.

Note that (82a) and (83a) are simplified due to the variable realization of the active prefix; see note 31 above.

6 Conclusion

Since the 1970s (see e.g. Keenan 1972, 1976a,b, Chung 1976a,b, Schachter 1976), western Austronesian languages have featured prominently in theoretical discussions around subjecthood and voice and their interactions with A'-movement. Many western Austronesian languages appear to exhibit a subject-only restriction on A'-extraction (Keenan and Comrie 1977 *et seq*; see also recent discussion in Branan and Erlewine 2024: 389–392), including the most well studied Malayic languages of Standard Indonesian (SI) and Standard Malay (SM): A'-extraction can only target the subject, as determined by voice morphology on the verb and reflected by its clause-initial position in basic word order, without changes in morphology. However, as first noted by Keenan (1972) and Chung (1976a) and elaborated upon by others (e.g. Saddy, 1991; Soh, 1998; Cole and Hermon, 2005), SI and SM *do* allow for the extraction of non-subject nominals if the verb appears exceptionally without its *meN*- active voice prefix. These facts have received significant interest in prior work, due to the apparent role that the shape of verbal morphology plays in the availability of syntactic operations.

In this paper, we developed a new proposal for the clausal morphosyntax of Malayic languages, which leads to a new and more explanatory account for the observed interactions between voice morphology and nominal A'-extraction. Central to our proposal is the assumed organization of the verbal domain, which draws especially on the prior proposal of Nomoto 2015, 2021 and synthesizes it with the idea that the verbal domain constitutes a cyclic domain — a *phase* — as in Aldridge 2008b, Sato 2012, among others. A lower functional head *v* introduces the external argument (agent) and a higher head Voice serves as the phase head, hosting exactly one nominal specifier. We further integrate Nomoto's proposal of these two functional heads with the idea that the Malayic active voice *meN*- reflects two prefixes (following Benjamin 2009, Gil 2002, and others; see section 5.1), associating Voice and *v* with *me*- and *N*- respectively.

The view of derivational cyclicity that we adopt then provides an explanatory link between the form of verbal affixes and the availability of certain movement configurations. Once a phase is built, the relative word order of its overt contents is then fixed, and cannot be contradicted by subsequent movement operations (Fox and Pesetsky, 2005). We showed that, with this view of phasehood, our proposal is able to derive the almost exceptionless subject-only restriction on nominal A'-movement in these languages, as well as the one exception of object extraction clauses and their verbal morphology. More concretely, what is important is the relationship between the external argument (e.g. agent, base-generated in Spec,*v*P) and the choice of morphology realized by the Voice head. Consider the A'-extraction of a nominal (DP_{A'}) that is not the local external argument via Spec,VoiceP as in (84a,b) below.

(84) **The external argument requires Voice to be unpronounced:**

	[CP	[TP	...	[VoiceP	Voice	[_v P	v+V
a. SI/SM-type object extr.:	DP _{A'}	DP _{ag}	Aux*	<i>t</i>	∅	<i>t</i>	∅-V <i>t</i>
b. Desa object extraction:	DP _{A'}		Aux*	<i>t</i>	∅	DP _{ag}	N-V <i>t</i>
c. Bare passive:		DP _{th}	Aux*	<i>t</i>	∅	DP _{ag}	∅-V <i>t</i>

For such a derivation to converge, one of two things must happen. The first possibility is that the external argument moves to the high subject position in Spec,TP, to proceed any auxiliaries (84a); this is possible only if Voice is unpronounced, making DP_{A'} and DP_{ag} the two leftmost parts of VoiceP, so that the relative word order established at the VoiceP phase is not later violated. The second possibility is that the external argument stays in-situ in Spec,vP, as in Desa object extraction (84b). If Voice were then realized as an overt prefix, the external argument would block its lowering onto the verb under linear adjacency with *v*, and so Voice must be unpronounced in this configuration as well.

This proposal, whereby the realization of Voice as overt or null underlies the availability of non-subject A'-extraction, offers a new understanding of the relationship between object extraction clauses and bare passives. Even though object extraction clauses importantly differ from bare passives — as reflected by the position of the subject in SI/SM and other varieties as in (84a) and by the form of the verb in Suak Mansi Desa as in (84b) — object extraction will only be grammatical if Voice is realized as a null allomorph, pruned from the representation and not subject to lowering or linearization. This requires the language to have a means of generating a null allomorph for Voice, as in the bare passive (see (84c)). The correlation between voice allomorph inventories and extraction possibilities in two registers of Madurese, among other facts, strongly supports this understanding of the relationship between bare passives and object extraction. More generally, our proposal highlights the important role that the choice of morphological forms can play in explaining syntactic phenomena, as is made possible through the Cyclic Linearization theory of phasehood (Fox and Pesetsky 2005, a.o.).

Finally, we note that our paper also stands apart from many prior works on the syntax of SI and SM in that we consider the broader range of attested voice and extraction interactions in regional Malayic languages and their varieties. In particular, we highlighted the behavior of Suak Mansi Desa, a previously undescribed Malayic language of West Kalimantan, in western Borneo, as particularly informative. While the morphosyntax of Desa at first glance appears quite similar to that of the standard languages, in particular syntactic configurations, their behaviors come apart. The consideration of such details in the grammars of

closely related languages directly motivated our proposal for the grammars of Malayic languages, including our new approach to the previously well-studied phenomenon of “*meN*-deletion” in SI and SM. This demonstrates the value of syntactic theory-building that is informed by a broader typological and comparative lens, including original empirical work on previously undescribed or marginalized varieties.

References

- Adelaar, Alexander. 2002. Salako morphology and the interrelation between voice, mood and aspect. In *Between worlds: Linguistic papers in memory of David John Prentice*, ed. Alexander Adelaar and Robert Blust, 1–28. Pacific Linguistics.
- Adelaar, Alexander. 2005a. Malayo-Sumbawan. *Oceanic Linguistics* 44:357–388.
- Adelaar, Alexander. 2005b. *Salako or Badameà: Sketch grammar, text, and lexicon of a Kanayatn dialect of West Borneo*. Harrassowitz Verlag.
- Adelaar, Alexander. 2005c. Structural diversity in the Malayic subgroup. In *The Austronesian languages of Asia and Madagascar*, ed. Alexander Adelaar and Nikolaus P. Himmelmann, 202–226. Routledge.
- Adelaar, Alexander. 2006. Where does Belangin belong? In *Insular Southeast Asia: Linguistic and cultural studies in honour of Bernd Nothofer*, ed. Fritz Schulze and Holger Warnk, 65–84. Otto Harrassowitz Verlag.
- Adelaar, K. Alexander. 1992. *Proto-Malayic: the reconstruction of its phonology and parts of its lexicon and morphology*. Pacific Linguistics.
- Aldridge, Edith. 2004. Ergativity and word order in Austronesian languages. Doctoral Dissertation, Cornell University.
- Aldridge, Edith. 2008a. Generative approaches to ergativity. *Language and Linguistics Compass* 10:966–995.
- Aldridge, Edith. 2008b. Phase-based account of extraction in Indonesian. *Lingua* 118:1440–1469.
- Amaechi, Mary, and Doreen Georgi. 2019. Quirks of subject (non-)extraction in Igbo. *Glossa* 4:1–36.
- Arka, Wayan, and Christopher D. Manning. 1998. Voice and grammatical relations in Indonesian: A new perspective. In *Proceedings of the LFG98 Conference*, ed. Miriam Butt and Tracy Holloway King, 45–69. CSLI Publications.
- Baker, Mark C. 1988. *Incorporation: A theory of grammatical function changing*. Chicago: University of Chicago Press.
- Barbosa, Pilar. 2001. On inversion in *wh*-questions in Romance. In *Subject inversion in Romance and the theory of Universal Grammar*, ed. Aafke Hulk and Jean-Yves Pollock, 20–59. Oxford University Press.
- Beavers, John, and I Nyoman Udayana. 2023. Middle voice as generalized argument suppression: The case from Indonesian. *Natural Language & Linguistic Theory* 41:51–102.
- Benjamin, Geoffrey. 2009. Affixes, Austronesian and iconicity in Malay. *Bijdragen tot de Taal-, Land- en Volkenkunde* 165:291–323.
- Blust, Robert. 2004. Austronesian nasal substitution: A survey. *Oceanic Linguistics* 43:73–148.
- Bobaljik, Jonathan David. 2000. The ins and outs of contextual allomorphy. In *University of Maryland working*

- papers in linguistics*, ed. Kleanthes K. Grohmann and Caro Struijke, volume 10, 35–71.
- Bobaljik, Jonathan David. 2017. Distributed morphology. In *Oxford research encyclopedia of linguistics*, ed. Mark Aronoff. Oxford University Press.
- Bossi, Madeline, and Michael Diercks. 2019. V1 in Kipsigis: Head movement and discourse-based scrambling. *Glossa* 4:1–43.
- Bošković, Željko. 2003. Agree, phases, and intervention effects. *Linguistic Analysis* 33:54–96.
- Bošković, Željko. 2007. On the locality and motivation of Move and Agree: An even more minimal theory. *Linguistic Inquiry* 38:589–644.
- Bošković, Željko. 2014. Now I’m a phase, now I’m not a phase: On the variability of phases with extraction and ellipsis. *Linguistic Inquiry* 45:27–89.
- Bošković, Željko. 2016. On the timing of labeling: Deducing Comp-trace effects, the Subject Condition, the Adjunct Condition, and tucking in from labeling. *The Linguistic Review* 33:17–66.
- Branan, Kenyon. 2023. Locality and antilocality: The logic of conflicting requirements. *Linguistic Inquiry* 54:1–38.
- Branan, Kenyon, and Michael Yoshitaka Erlewine. 2024. $\bar{A}$ -probing for the closest DP. *Linguistic Inquiry* 55:375–401.
- Brodkin, Dan. 2022. Two steps to high absolutive syntax: Austronesian voice and Agent Focus in Mandar. *Journal of East Asian Linguistics* 31:465–516.
- Brodkin, Dan. 2025. Subject positions, high absolutives, and Austronesian voice. University of California Santa Cruz, ms.
- Bruening, Benjamin. 2010. Language-particular syntactic rules and constraints: English locative inversion and *do*-support. *Language* 86:43–84.
- Carstens, Vicki. 2005. Agree and EPP in Bantu. *Natural Language & Linguistic Theory* 23:219–279.
- Chomsky, Noam. 1973. Conditions on transformations. In *A festschrift for Morris Halle*, ed. Stephen R. Anderson and Paul Kiparsky. New York: Holt, Reinhart, and Winston.
- Chomsky, Noam. 1977. On *wh*-movement. In *Formal syntax*, ed. Peter Culicover, Thomas Wasow, and Adrian Akmajian, 71–132. New York: Academic Press.
- Chomsky, Noam. 1980. On binding. *Linguistic Inquiry* 11:1–46.
- Chomsky, Noam. 1981. *Lectures on government and binding*. Foris.
- Chomsky, Noam. 1982. *Some concepts and consequences of the theory of government and binding*. MIT Press.
- Chomsky, Noam. 1995. *The minimalist program*. MIT Press.

- Chomsky, Noam. 2000. Minimalist inquiries: The framework. In *Step by step: Essays on minimalist syntax in honor of Howard Lasnik*, ed. Roger Martin, David Michaels, and Juan Uriagereka, 89–156. MIT Press.
- Chomsky, Noam. 2001. Derivation by phase. In *Ken Hale: A life in language*, ed. Michael Kenstowicz, 1–52. MIT Press.
- Chung, Sandra. 1976a. An object-creating rule in Bahasa Indonesia. *Linguistic Inquiry* 7:41–87.
- Chung, Sandra. 1976b. On the subject of two passives in Indonesian. In Li (1976), 57–99.
- Chung, Sandra. 1978. Stem sentences in Indonesian. In *Second International Conference on Austronesian Linguistics: Proceedings*, ed. Stephen A. Wurm and Lois Carrington, 335–365. Pacific Linguistics.
- Chung, Sandra. 1982. Unbounded dependencies in Chamorro grammar. *Linguistic Inquiry* 13:39–78.
- Chung, Sandra. 1998. *The design of agreement: Evidence from Chamorro*. University of Chicago Press.
- Chung, Sandra, and Maria Polinsky. 2009. Introduction to the special issue on comparative syntax of Austronesian languages. *Natural Language & Linguistic Theory* 27:659–673.
- Cole, Peter, and Gabriella Hermon. 1998. The typology of *wh*-movement: *wh*-questions in Malay. *Syntax* 1:221–258.
- Cole, Peter, and Gabriella Hermon. 2000. Partial *wh*-movement: Evidence from Malay. In *Wh-scope marking*, ed. Uli Lutz, Gereon Müller, and Arnim von Stechow, 101–130. John Benjamins.
- Cole, Peter, and Gabriella Hermon. 2005. Subject and non-subject relativization in Indonesian. *Journal of East Asian Linguistics* 14:59–88.
- Cole, Peter, Gabriella Hermon, and Norhaida Bte. Aman. 1999a. Clefted questions in Malay. University of Delaware, ms.
- Cole, Peter, Gabriella Hermon, and Yassir Tjung. 2005. How irregular is *wh* in situ in Indonesian? *Studies in Language* 29:553–581.
- Cole, Peter, Gabriella Hermon, and Yassir Tjung. 2006. Is there *pasif semu* in Jakarta Indonesian? *Oceanic Linguistics* 45:64–90.
- Cole, Peter, Gabriella Hermon, and Yanti. 2008. Voice in Malay/Indonesian. *Lingua* 118:1500–1553.
- Cole, Peter, Elizabeth Jonczyk, and Jason Lilley. 1999b. A note on extraction from object position in Javanese and other Javanic languages. In *Proceedings of AFLA 6*, ed. Carolyn Smallwood and Catherine Kitto, 87–93. Toronto Working Papers in Linguistics.
- Collins, Chris. 2005. A smuggling approach to the passive in English. *Syntax* 8:81–120.
- Collins, James T. 2004. Ibanic languages in Kalimantan Barat, Indonesia: Exploring nomenclature, distribution, and characteristics. *Borneo Research Bulletin* 35:17–47.
- Collins, James T., and Herpanus. 2018. The Sekujam language of West Kalimantan (Indonesia). *Wacana*

- 19:425–458.
- Coon, Jessica, Pedro Mateo Pedro, and Omer Preminger. 2014. The role of case in A-bar extraction asymmetries: Evidence from Mayan. *Linguistic Variation* 14:179–242.
- Dardjowidjojo, Soenjono. 1978. *Sentence patterns of Indonesian*. University of Hawai‘i Press.
- Davies, William D. 2010. *A grammar of Madurese*. Mouton.
- Davis, Colin. 2020a. Crossing and stranding at edges: On intermediate stranding and phase theory. *Glossa: a journal of general linguistics* 5:1–32.
- Davis, Colin. 2020b. The linear limitations of syntactic derivations. Doctoral Dissertation, Massachusetts Institute of Technology.
- Davis, Colin. 2021. Possessor extraction in colloquial English: Evidence for successive cyclicity and cyclic linearization. *Linguistic Inquiry* 52:291–332.
- Deal, Amy Rose. 2016. Syntactic ergativity: Analysis and identification. *Annual Review of Linguistics* 2:1–21.
- Deal, Amy Rose. 2019. Raising to ergative: Remarks on applicatives of unaccusatives. *Linguistic Inquiry* 50:388–415.
- Dempwolff. 1934. *Vergleichende lautlehre des austronesischen Wortschatzes, volume 1: Induktiver Aufbau einer indonesischen Ursprache*.
- Denistia, Karlina. 2018. Revisiting the Indonesian prefixes *pen-*, *pe₂-*, and *per-*. *Linguistik Indonesia* 36:144–160.
- Denistia, Karlina, and R. Harald Baayen. 2019. The Indonesian prefixes *pe-* and *pen-*: A study in productivity and allomorphy. *Morphology* 29:385–407.
- Dixon, R. M. W. 1994. *Ergativity*. Cambridge University Press.
- Donohue, Mark. 2008. Malay as a mirror of Austronesian: Voice development and voice variation. *Lingua* 118:1470–1499.
- Drummond, Emily. 2022. Abstract ergative Case without morphological case. In *Proceedings of NELS 52*, 259–272.
- Drummond, Emily. 2023. Clause structure and ergativity in Nukuoro. Doctoral Dissertation, University of California Berkeley.
- Dunselman, P. Donatus. 1949. Bijdragen tot de kennis van de taal en adat der Kendajan Dajaks van West-Borneo. *Bijdragen tot de Taal-, Land- en Volkenkunde* 105:59–105.
- Embick, David. 2003. Locality, listedness, and morphological identity. *Studia Linguistica* 57:143–169.
- Embick, David. 2010. *Localism versus globalism in morphology and phonology*. MIT Press.

- Embick, David, and Rolf Noyer. 2001. Movement operations after syntax. *Linguistic Inquiry* 32:555–595.
- Embick, David, and Rolf Noyer. 2007. Distributed Morphology and the syntax/morphology interface. In *The Oxford handbook of linguistic interfaces*, ed. Gillian Ramchand and Charles Reiss, 289–324. Oxford University Press.
- Erlewine, Michael Yoshitaka. 2016a. Anti-locality and optimality in Kaqchikel Agent Focus. *Natural Language & Linguistic Theory* 34:429–479.
- Erlewine, Michael Yoshitaka. 2016b. Review of *Minimalist Interfaces: Evidence from Indonesian and Javanese* by Yosuke Sato. *Oceanic Linguistics* 55:298–306.
- Erlewine, Michael Yoshitaka. 2017. Why the null complementizer is special in complementizer-trace effects. In *A pesky set: Papers for David Pesetsky*, ed. Claire Halpert, Hadas Kotek, and Coppe van Urk, 371–380. MIT Working Papers in Linguistics.
- Erlewine, Michael Yoshitaka. 2018. Extraction and licensing in Toba Batak. *Language* 94:662–697.
- Erlewine, Michael Yoshitaka. 2020. Anti-locality and subject extraction. *Glossa* 5:1–38.
- Erlewine, Michael Yoshitaka, Theodore Levin, and Coppe van Urk. 2020. The typology of nominal licensing in Austronesian voice system languages. In *Proceedings of AFLA 26*, ed. Ileana Paul, 71–87.
- Fortin, Catherine, and Hooi Ling Soh. 2014. Blocking effects and the verbal prefix *ber-* in Malay and Indonesian. In *Proceedings of AFLA 20*.
- Fortin, Catherine Rose. 2007. Indonesian sluicing and verb phrase ellipsis: Description and explanation in a minimalist framework. Doctoral Dissertation, University of Michigan.
- Fox, Danny, and David Pesetsky. 2005. Cyclic linearization of syntactic structure. *Theoretical Linguistics* 31:1–45.
- Gallego, Ángel. 2008. Four reasons to push down the external argument. Universitat Autònoma de Barcelona, MS.
- Georgi, Doreen. 2014. Opaque interactions of Merge and Agree: On the nature and order of elementary operations. Doctoral Dissertation, Universität Leipzig.
- Georgi, Doreen. 2017. Patterns of movement reflexes as the result of the order of Merge and Agree. *Linguistic Inquiry* 48:585–626.
- Gil, David. 2002. The prefixes *di-* and *N-* in Malay/Indonesian dialects. In *The history and typology of western Austronesian voice systems*, ed. Fay Wouk and Malcolm Ross, 241–283. Canberra: Pacific Linguistics.
- Guilfoyle, Eithne, Henrietta Hung, and Lisa Travis. 1992. Spec of IP and Spec of VP: Two subjects in Austronesian languages. *Natural Language & Linguistic Theory* 10:375–414.
- Harley, Heidi, and Rolf Noyer. 1999. Distributed morphology. *Glott International* 4:3–9.

- Hassan, Abdullah. 1974. *The morphology of Malay*. Kuala Lumpur: Dewan Bahasa dan Pustaka.
- Herpanus. 2014. *Bahasa dan etnobotani suku Desa: Tantangan perubahan ekologi*. Pontianak: STAIN Pontianak Press.
- Himmelman, Nikolaus P. 2005. The Austronesian languages of Asia and Madagascar: Typological characteristics. In *The Austronesian languages of Asia and Madagascar*, ed. Alexander Adelaar and Nikolaus P. Himmelman, 110–181. Routledge.
- Holmberg, Anders. 2000. Scandinavian stylistic fronting: How any category can become an expletive. *Linguistic Inquiry* 31:445–483.
- Hsieh, Henrison. 2020. Beyond nominative: A broader view of $\bar{A}$ -dependencies in Tagalog. Doctoral Dissertation, McGill University.
- Hsieh, Henrison. 2025. Locality in exceptional Tagalog A'-extraction. *Linguistic Inquiry* 56:519–560.
- Jeoung, Helen. 2016. Voice and register differences in Madurese. In *Proceedings of AFLA 22*, ed. Henrison Hsieh, 49–65. Asia-Pacific Linguistics.
- Jeoung, Helen. 2017. On the number of voices in Madurese. *Journal of the Southeast Asian Linguistics Society* 10:16–35.
- Jeoung, Helen. 2018a. Optional elements in Indonesian morphosyntax. Doctoral Dissertation, University of Pennsylvania.
- Jeoung, Helen. 2018b. Possessors move through the edge, too. *Glossa* 3:1–35.
- Jouitteau, Mélanie. 2007. The Brythonic reconciliation. *Linguistic Variation Yearbook* 7:163–200.
- Kader, Mashudi. 1976. The syntax of Malay interrogatives. Doctoral Dissertation, Simon Fraser University.
- Kalom, Ina Anak, and Alfred B. Hudson. 1970. Selako traditional history: a story on the origins of Kampong Pueh. *Sarawak Museum Journal* 18:281–300.
- Karaj, David M., and Andrea Sansò. 2023. From verbalizer to middle marker: the diachrony of middle voice in Malayic. *STUF - Language Typology and Universals* 76:217–234.
- Kaswanti Purwo, Bambang. 1986. The presence and absence of *men-*: the Indonesian transitive verbs. In *FOCAL i: Papers from the fourth international conference on Austronesian linguistics*, ed. Paul Geraghty, Lois Carrington, and Stephen A. Wurm, 159–170. Pacific Linguistics.
- Kaufman, Daniel. 2007. Review of Adelaar 2005 *Salako or Badameà, sketch grammar, texts and lexicon of a Kanayatn dialect in West Borneo*. *Oceanic Linguistics* 46:624–633.
- Kaufman, Daniel. 2014. The syntax of Indonesian imposters. In *Cross-linguistic studies of imposters and pronominal agreement*, ed. Chris Collins, 89–120. Oxford University Press.
- Keenan, Edward. 1976a. Remarkable subjects in Malagasy. In Li (1976), 247–301.

- Keenan, Edward L. 1972. Relative clause formation in Malagasy. In *The Chicago which hunt*, ed. Paul M. Peranteau, Judith N. Levi, and Gloria C. Phares, 169–189. Chicago Linguistic Society.
- Keenan, Edward L. 1976b. Towards a universal definition of “subject”. In Li (1976), 303–333.
- Keenan, Edward L., and Bernard Comrie. 1977. Noun phrase accessibility and Universal Grammar. *Linguistic Inquiry* 8:63–99.
- Keine, Stefan, and Hedde Zeijlstra. 2025. Clause-internal successive cyclicity: Phasality or DP intervention? *Natural Language & Linguistic Theory* 43:1119–1182.
- Kemmer, Suzanne. 1993. *The middle voice*. John Benjamins.
- Kenstowicz, Michael. 1983. The null-subject parameter in modern Arabic dialects. In *Proceedings of NELS 14*, ed. Charles Jones and Peter Sells, 207–219.
- Kenstowicz, Michael. 1989. The null subject parameter in modern Arabic dialects. In *The null subject parameter*, ed. Osvaldo Jaeggli and Ken Safir, 263–275. Kluwer.
- Kinyalolo, Kasangati Kikuni Wagongambilu. 1991. Syntactic dependencies and the Spec-Head agreement hypothesis in KiLega. Doctoral Dissertation, University of California at Los Angeles.
- Kitagawa, Yoshihisa. 1986. Subjects in Japanese and English. Doctoral Dissertation, University of Massachusetts Amherst.
- Ko, Heejeong. 2007. Asymmetries in scrambling and cyclic linearization. *Linguistic Inquiry* 38:49–83.
- Ko, Heejeong. 2011. Predication and edge effects. *Natural Language & Linguistic Theory* 29:725–778.
- Ko, Heejeong. 2014. *Edges in syntax: Scrambling and Cyclic Linearization*. Oxford University Press.
- Kratzer, Angelika. 1996. Severing the external argument from its verb. In *Phrase structure and the lexicon*, ed. Johan Rooryck and Laurie Zaring, 109–137. Springer.
- Kroeger, Paul R. 1991. Phrase structure and grammatical relations in Tagalog. Doctoral Dissertation, Stanford University. Reprinted by CSLI Publications, 1993.
- Kroeger, Paul R. 2009. Malagasy clefts from a Western Malayo-Polynesian perspective: Commentary on the paper by Hans-Martin Gärtner. *Natural Language & Linguistic Theory* 27:817–838.
- Kuroda, Sige-Yuki. 1988. Whether we agree or not: A comparative syntax of English and Japanese. *Linguisticae Investigations* 12:1–47.
- Lapoliwa, Hans. 1981. A generative approach to the phonology of Bahasa Indonesia. Doctoral Dissertation, SOAS University of London.
- Lee, Tommy Tsz-Ming. 2021. Asymmetries in doubling and Cyclic Linearization. *Journal of East Asian Linguistics* 30:109–139.
- Legate, Julie Anne. 2008. Morphological and abstract case. *Linguistic Inquiry* 39:55–101.

- Legate, Julie Anne. 2011. Under-inheritance. Presented at NELS 42.
- Legate, Julie Anne. 2012a. Subjects in Acehnese and the nature of the passive. *Language* 88:495–525.
- Legate, Julie Anne. 2012b. Types of ergativity. *Lingua* 122:181–191.
- Legate, Julie Anne. 2014. *Voice and v: Lessons from Acehnese*. MIT Press.
- Levin, Theodore. 2015. Licensing without case. Doctoral Dissertation, Massachusetts Institute of Technology.
- Li, Charles N., ed. 1976. *Subject and topic*. Academic Press.
- MacDonald, R. Ross, and Soenjono Darjowidjojo. 1967. *A student's reference grammar of modern formal Indonesian*. Georgetown University Press.
- Maienborn, Claudia. 2020. Revisiting Olga, the beautiful dancer: An intersective A-analysis. In *Proceedings of SALT 30*, 63–82.
- Manning, Christopher D. 1996. *Ergativity: Argument structure and grammatical relations*. CSLI Publications.
- Marantz, Alec. 1988. Clitics, morphological merger, and the mapping to phonological structure. In *Theoretical morphology*, 253–270. Academic Press.
- McCarthy, John J., and Alan S. Prince. 1995. Faithfulness and reduplicative identity. In *Papers in optimality theory*, ed. Jill Beckman, Suzanne Urbanczyk, and Laura Walsh Dickey, volume 18 of *University of Massachusetts Occasional Papers in Linguistics*.
- McCloskey, James. 1979. *Transformational syntax and model theoretic semantics: A case study in Modern Irish*. Dordrecht: Reidel.
- McCloskey, James. 2001. The morphosyntax of *wh*-extraction in Irish. *Journal of Linguistics* 37:67–100.
- Medeiros, David J. 2013. Hawaiian VP-remnant movement: A cyclic linearization approach. *Lingua* 127:72–97.
- Merchant, Jason. 2013. Voice and ellipsis. *Linguistic Inquiry* 44:77–108.
- Musgrave, Simon. 2001. Non-subject arguments in Indonesian. Doctoral Dissertation, University of Melbourne.
- Nomoto, Hiroki. 2006. Voice in colloquial Malay relatives. In *Gengojouhougaku kenkyuuhoukoku*, ed. Yoichiro Tsuruga, Nobuyuki Miyake, and Yuji Kawaguchi, 12, 97–116. Tokyo University of Foreign Studies.
- Nomoto, Hiroki. 2010. Making sense of the optionality of voice marking in Malay/Indonesian. In *Proceedings of the Workshop on Indonesian-type Voice System*, 37–44. Tokyo University of Foreign Studies.
- Nomoto, Hiroki. 2013. On the optionality of grammatical markers: A case study of voice marking in Malay/Indonesian. *NUSA: Linguistic studies of languages in and around Indonesia* 54:121–143.
- Nomoto, Hiroki. 2015. Person restriction on passive agents in Malay and givenness. In *Proceedings of the*

- Second International Workshop on Information Structure of Austronesian Languages*, 83–101.
- Nomoto, Hiroki. 2016. Passives and clitic doubling: A view from Classical Malay. In *Proceedings of AFLA 23*, ed. Hiroki Nomoto, Takuya Miyauchi, Asako Shiohara, and Takuya Miyauchi, 179–193. Asia-Pacific Linguistics.
- Nomoto, Hiroki. 2020. Passive subtypes in Sarawak Malay. *NUSA: Linguistic studies of languages in and around Indonesia* 68:139–158.
- Nomoto, Hiroki. 2021. Bare passive agent hierarchy. In *Proceedings of AFLA 27*, ed. Henrison Hsieh and Keely New, 57–70.
- Onn, Farid. 1976. Aspects of Malay phonology and morphology: a generative approach. Doctoral Dissertation, University of Illinois at Urbana-Champaign.
- Perlmutter, David M. 1968. Deep and surface constraints on syntax. Doctoral Dissertation, Massachusetts Institute of Technology.
- Pesetsky, David. 2017. Complementizer-trace effects. In *Blackwell Companion to Syntax*, ed. Martin Everaert and Henk van Riemsdijk. Wiley-Blackwell, second edition.
- Pietraszko, Asia. 2023. Timing-driven derivation of a NOM/ACC agreement pattern. *Glossa* 8:1–55.
- Polinsky, Maria. 2016. *Deconstructing ergativity: Two types of ergative languages and their features*. Oxford University Press.
- Polinsky, Maria. 2017. Syntactic ergativity. In *Blackwell Companion to Syntax*, ed. Martin Everaert and Henk van Riemsdijk. Wiley-Blackwell, second edition.
- Postal, Paul M. 2004. *Skeptical linguistic essays*. Oxford University Press.
- Potsdam, Eric. 2009. Austronesian verb-initial languages and *wh*-question strategies. *Natural Language & Linguistic Theory* 27:737–771.
- Richards, Norvin. 2010. *Uttering trees*. MIT Press.
- Richards, Norvin. 2023. Two kinds of bans on ergative extraction. MIT, ms.
- Ross, Malcolm. 2002. The history and transitivity of western Austronesian voice and voice marking. In *The history and typology of western Austronesian voice systems*, ed. Fay Wouk and Malcolm Ross, 17–62. Canberra: Pacific Linguistics.
- Sabbagh, Joseph. 2007. Ordering and linearizing rightward movement. *Natural Language & Linguistic Theory* 25:349–401.
- Saddy, Doug. 1991. *Wh*-scope mechanisms in Bahasa Indonesia. In *More papers on wh-movement*, ed. Lisa Lai-Shen Cheng and Hamida Demirdache, 183–218. MIT Working Papers in Linguistics.
- Saito, Mamoru. 1985. Some asymmetries in Japanese and their theoretical implications. Doctoral Dissertation,

- Massachusetts Institute of Technology.
- Sato, Yosuke. 2012. Successive cyclicity at the syntax-morphology interface: Evidence from Standard Indonesian and Kendal Javanese. *Studia Linguistica* 66:32–57.
- Schachter, Paul. 1976. The subject in Philippine languages: Topic, actor, actor-topic, or none of the above. In Li (1976), 491–518.
- Sheehan, Michelle. 2018. Nominal licensing in caseless languages. *Journal of Linguistics* 54:527–589.
- Sigurðsson, Halldór Ármann. 2010. On EPP effects. *Studia Linguistica* 64:159–238.
- Smith, Alexander D. 2017. The languages of Borneo: A comprehensive classification. Doctoral Dissertation, University of Hawai‘i.
- Sneddon, James Neil. 1996. *Indonesian: A comprehensive grammar*. Routledge.
- Soh, Hooi Ling. 1998. Certain restrictions on A-bar movement in Malay. In *Proceedings of the third and fourth meetings of Austronesian Formal Linguistics Association (AFLA)*, ed. Matthew Pearson, number 21 in UCLA Occasional Papers in Linguistics, 295–307. Los Angeles: Department of Linguistics, UCLA.
- Soh, Hooi Ling. 2013. Voice and aspect: Some notes on Malaysian Malay. *NUSA: Linguistic studies of languages in and around Indonesia* 54:159–173.
- Soh, Hooi Ling, and Hiroki Nomoto. 2009. Progressive aspect, the verbal prefix *meN-*, and stative sentences in Malay. *Oceanic Linguistics* 48:148–171.
- Soh, Hooi Ling, and Hiroki Nomoto. 2015. Degree achievements, telicity, and the verbal prefix *meN-* in Malay. *Journal of Linguistics* 51:147–183.
- Sommerlot, Carly J. 2020. On the syntax of West Kalimantan: Asymmetries and A’-movement in Malayic and Land Dayak languages. Doctoral Dissertation, University of Texas at Arlington.
- Sommerlot, Carly J. 2023. Languages of West Kalimantan: Desa. Kaipuleohone Digital Language Archive. URL <https://hdl.handle.net/10125/104698>.
- Sommerlot, Carly J. 2024. Morphosyntactic variation in the Kendayan branch. National University of Singapore, ms.
- Stevens, Alan M. 1970. Pseudo-transitive verbs in Indonesian. *Indonesia* 9:67–72.
- Svenonius, Peter. 2002. Introduction. In *Subjects, expletives, and the EPP*, ed. Peter Svenonius, 3–28. Oxford University Press.
- Tharp, James A. 1974. Phonological processes associated with nasal verbal prefixes in Western Austronesian languages. ms.
- Tjung, Yassir. 2006. The formation of relative clauses in Jakarta Indonesian: A subject-object asymmetry. Doctoral Dissertation, University of Delaware.

- Vamarasi, Marit Kana. 1999. *Grammatical relations in Bahasa Indonesia*. Pacific Linguistics.
- van Urk, Coppe. 2020. Successive cyclicity and the syntax of long-distance dependencies. *Annual Review of Linguistics* 6:111–130.
- van Urk, Coppe, and Norvin Richards. 2015. Two components of long-distance extraction: Successive cyclicity in Dinka. *Linguistic Inquiry* 46:113–155.
- Vergnaud, Jean-Roger. 1977. Letter to Noam Chomsky and Howard Lasnik. In *Reprinted in foundational issues in linguistic theory: essays in honor of Jean-Roger Vergnaud*, ed. Robert Freidin, Carlos P. Otero, and Maria-Luisa Zubizarreta, 2008, 3–15. MIT Press.
- Voskuil, Jan E. 1996. Aspects of verbal morphology in Tagalog. *NUSA: Linguistic studies of languages in and around Indonesia* 39:51–66.
- Voskuil, Jan E. 2000. Indonesian voice and A-bar movement. In *Formal issues in Austronesian linguistics*, ed. Ileana Paul, Vivianne Phillips, and Lisa Travis, 195–213. Springer.
- Wee, Hock Ann Lionel. 1995. Cognition in grammar: The problem of verbal prefixation in Malay. Doctoral Dissertation, University of California Berkeley.
- Wolff, John U. 1996. The development of the passive verb with pronominal prefix in western Austronesian languages. In *Reconstruction, classification, description: Festschrift in honor of Isidore Dyen*, ed. Bernd Nothofer, 15–40. Abera.
- Wong, Wei Chen. 1998. The middle voice in Malay. Master's thesis, National University of Singapore.
- Woolford, Ellen. 2006. Lexical case, inherent case, and argument structure. *Linguistic Inquiry* 37:111–130.
- Yang, Charles. 2016. *The price of productivity*. MIT Press.
- Yanti. 2010. A reference grammar of Jambi Malay. Doctoral Dissertation, University of Delaware.
- Zobel, Sarah. 2017. The sensitivity of natural language to the distinction between class nouns and role nouns. In *Proceedings of SALT 27*, 438–458.